

LLMs Are Few-Shot In-Context Low-Resource Language Learners

Samuel Cahyawijaya
HKUST

scahyawijaya@connect.ust.hk holy@aisingapore.org

Holy Lovenia
AI Singapore

Pascale Fung
HKUST

pascale@ece.ust.hk

Abstract

In-context learning (ICL) empowers large language models (LLMs) to perform diverse tasks in underrepresented languages using only short in-context information, offering a crucial avenue for narrowing the gap between high-resource and low-resource languages. Nonetheless, there is only a handful of works explored ICL for low-resource languages with most of them focusing on relatively high-resource languages, such as French and Spanish. In this work, we extensively study ICL and its cross-lingual variation (X-ICL) on 25 low-resource and 7 relatively higher-resource languages. Our study not only assesses the effectiveness of ICL with LLMs in low-resource languages but also identifies the shortcomings of in-context label alignment, and introduces a more effective alternative: query alignment. Moreover, we provide valuable insights into various facets of ICL for low-resource languages. Our study concludes the significance of few-shot in-context information on enhancing the low-resource understanding quality of LLMs through semantically relevant information by closing the language gap in the target language and aligning the semantics between the targeted low-resource and the high-resource language that the model is proficient in. Our work highlights the importance of advancing ICL research, particularly for low-resource languages. Our code is publicly released at <https://github.com/SamuelCahyawijaya/in-context-alignment>.

1 Introduction

Large language models (LLMs) have displayed remarkable generalization capability in various tasks (Brown et al., 2020b; Kojima et al., 2022; Wei et al., 2022; Smith et al., 2022; Rae et al., 2022; Chowdhery et al., 2022; Scao et al., 2022; Liang et al., 2023; Srivastava et al., 2023; Lovenia et al., 2023; Bang et al., 2023). Nonethe-

less, these models face difficulties in generalizing across different languages, leading to performance disparity, particularly for low-resource languages (Aji et al., 2022a; Ebrahimi et al., 2022a; Adelani et al., 2022b; Cahyawijaya et al., 2023a,b; Asai et al., 2023). A myriad of research works address this problem through language-specific fine-tuning (Wilie et al., 2020; Kakwani et al., 2020; Cahyawijaya et al., 2021; Adelani et al., 2021; Kumar et al., 2022) which often leads to catastrophic forgetting (French, 1993; Chaudhry et al., 2019; Rolnick et al., 2019). Another line of work utilizes continual learning and adapter-based methods to inject new languages to existing LLMs (Yong et al., 2022; Cahyawijaya et al., 2023c; Jin et al., 2023). Nevertheless, these methods rely on performing multiple steps of parameter updates which require huge computational budgets, particularly for very large LLMs with hundreds of billion parameters.

To cope with this problem, prior works (Winata et al., 2021b; Lin et al., 2022a; Shi et al., 2023; Zhang et al., 2023) explore cross-lingual in-context learning (X-ICL) methods, an extension from in-context learning (ICL), that allow LLMs to generate better response quality in low-resource languages without the need for parameter tuning. In X-ICL, source language exemplars are incorporated into the input context allowing the model to transfer the task understanding capability from the source, commonly high-resource, language into the target language query (Winata et al., 2021b; Shi et al., 2023). However, X-ICL still fails to compete with a simple translate-test baseline, prominently for low-resource languages. A recent work (Tanwar et al., 2023) further enhances X-ICL through semantically similar cross-lingual exemplars and in-context label alignment¹, yielding a large gain over the baselines on relatively high-resource languages

¹In Tanwar et al. (2023), label alignment is referred to as task alignment. In this work, we distinguish two types of task alignments, i.e., query alignment and label alignment (§3.1).

such as French, Spanish, Chinese, and Japanese.

In this work, we expand upon the concept of cross-lingual semantic similarity and in-context alignment, specifically focusing on low-resource languages. Our hypothesis posits that their effectiveness may be compromised in low-resource languages due to the weak representation of the labels and sentences for the target languages. To test our hypothesis, we explore cross-lingual in-context learning (X-ICL) covering 25 low-resource languages from various language families and compare them with the performance of 7 relatively higher resource languages, including French (fra), Spanish (spa), German (deu), Italian (ita), Portuguese (por), Arabic (arb), and Hindi (hin). Our result suggests that the X-ICL performance decays correlate to the size of pre-training data of the target languages, which aligns with our hypothesis. Moreover, to our surprise, contrary to the results reported in (Tanwar et al., 2023), we found that in-context label alignment does not work for all the languages under study and introduced an alternative alignment method namely in-context query alignment, which significantly improves the alignment quality compared to the in-context label alignment.

To this end, we explore alternatives for X-ICL approaches covering variations of in-context alignment information, prompt, label encoding, and strategy for selecting in-context learning exemplars. We extensively analyze all these factors and their effect on the downstream task performance of all the languages under study. Our results and analysis highlight the following key takeaways:

- Contrary to prior work (Tanwar et al., 2023), we found that label alignment undermines the performance in most languages. Keeping uniform labels from the high-resource language often yields the best results.
- We introduce a new approach for cross-lingual alignment, i.e., query alignment, which is more effective than label alignment and can substitute or complement X-ICL.
- We analyze the effect of improving prompt format consistency on low-resource languages. However, despite improving performance for higher-resource languages, format consistency does not yield any benefit to the low-resource languages under study.
- We present a comprehensive in-context learning framework for better low-resource language understanding under various con-

strained conditions, concluding the significance of few-shot in-context information on enhancing the low-resource understanding quality of LLMs through semantically relevant information, where **monolingual ICL** does so by closing the language and domain gap on the targeted downstream task, while **X-ICL** closes the domain gap to the target downstream task, and **in-context alignment** closes the semantic gap between the targeted low-resource and the high-resource language that the model is proficient in.

2 Related Work

2.1 In-Context Learning

The in-context learning paradigm, originally introduced by Brown et al. (2020a), has significantly advanced our understanding of LLMs’ capabilities. It demonstrated that LLMs can effectively perform complex tasks through in-context learning with just task-specific formatting and a few task-specific examples (few-shot) or none at all (zero-shot). This ability is facilitated by the LLMs’ increasing capacity for generalization across diverse tasks, e.g., machine translation, question answering, and domain adaptation, without gradient updates.

Another line of work expands the scope of the study to multilingual generative LLMs, i.e., BLOOM (Scao et al., 2022), trained on the ROOTS corpus covering 46 natural and 13 programming languages, and XGLM (Lin et al., 2022b), trained on 500B tokens comprising 30 languages, which exhibit robust zero-shot and few-shot performances on multilingual NLP tasks. Furthermore, Lin et al. (2022b) address the imbalance in language representation by up-sampling the less-resourced languages. Bandarkar et al. (2023) then expand the language coverage of the in-context learning evaluation to 122 languages through Belebele, a wide-scale multilingual multiple-choice machine reading comprehension benchmark comprising short passages from FLORES-200 (Goyal et al., 2022).

Cross-Lingual In-Context Learning (X-ICL) Winata et al. (2021b) were among the first to explore the potential of few-shot X-ICL. Using 4 high-resource languages, this work shows that pre-trained LMs significantly outperform random prediction in cross-lingual tasks and produce better results compared to smaller fine-tuned baselines. Winata et al. (2022a) expand this study to un-

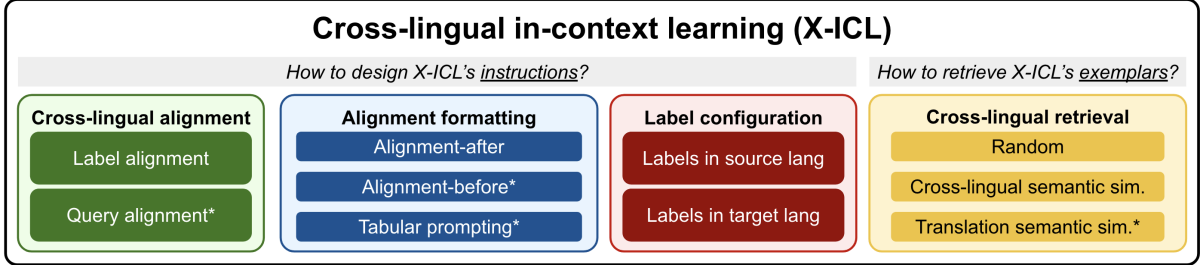


Figure 1: Framework of cross-lingual in-context learning (X-ICL) methods analyzed in our work. We explore various cross-lingual retrieval methods with different kinds of cross-lingual prompting strategies. *Novel approaches.

seen languages and find that taking the X-ICL contexts from a mixture of random source languages is surprisingly more effective compared to linguistically similar and geographically similar languages. Expanding the investigation to different aspects of cross-lingual transfers in X-ICL, Lin et al. (2022b) explores the use of various languages for instructions and exemplars. They find that incorporating English instructions notably improves zero-shot performance across multiple languages.

In a related vein, Tanwar et al. (2023) analyze the effect of cross-lingual prompt design for X-ICL across 3 text classification tasks using 44 different cross-lingual pairs. Their findings emphasize the limitations of random exemplar selection and propose the use of semantic-based exemplar retrieval and label alignment¹ for superior X-ICL performance. Notably, their findings diverge from our results (§5.1), which contend that label alignment does not provide benefits for X-ICL.

2.2 LLMs on Low-Resource Languages

Rigorous evaluations have been proposed to investigate how LLMs perform on low-resource languages. According to (Cahyawijaya et al., 2023a), while multilingual LLMs typically exhibit positive transfer learning among related languages, these models perform notably better for mid- and high-resource (e.g., Indonesian and English) compared to low-resource languages (e.g., other 18 Indonesian indigenous languages). This implies a challenge in the generalization capability of existing multilingual LLMs to low-resource languages. This is further evidenced by (Cahyawijaya et al., 2023b), which extends the exploration to 12 low-resource and extremely low-resource languages, where both existing zero-shot prompting LLMs and fine-tuned pre-trained LMs struggle to outperform classical machine learning baselines, which is indicative of LLMs’ limited ability to generalize to extremely low-resource languages that are

significantly distinct from those encountered during their training. Similar observations have been reported by Asai et al. (2023); Bang et al. (2023); Adilazuarda et al. (2024) for lower-resource languages. Furthermore, another line of work emphasizes the challenges faced by multilingual LLMs in understanding (Zhang et al., 2023; Adilazuarda et al., 2023) and generating (Yong et al., 2023) code-switching, a real use case and nuance of multilingualism exhibited by human speakers.

3 Methods

Figure 1 presents the general framework of X-ICL, comprising: 1) cross-lingual in-context alignment (§3.1), 2) cross-lingual alignment formatting and 3) label configuration as parts of cross-lingual prompting (§3.2), as well as 4) cross-lingual retrieval (§3.3). We assess variations of these X-ICL aspects to understand their effectiveness on different language resource levels.

3.1 Cross-Lingual Alignment

Prior works showcase the benefit of cross-lingual in-context learning with random exemplars which can improve the zero-shot performance of LLMs on downstream tasks (Winata et al., 2021c; Shi et al., 2023; Asai et al., 2023). More recently, Tanwar et al. (2023) introduce cross-lingual in-context alignment that injects a label aligner to the prompt in between the in-context exemplars and the input query. The label aligner provides the translation of the source label set $C^{src} = \{c_1^{src}, c_2^{src}, \dots, c_k^{src}\}$ to the target label set $C^{tgt} = \{c_1^{tgt}, c_2^{tgt}, \dots, c_k^{tgt}\}$. For instance, given a target language L^{tgt} , the label aligner prompt is formatted as follow: “In L^{tgt} , c_1^{src} means c_1^{tgt} , c_2^{src} means c_2^{tgt} , ..., and c_k^{src} means c_k^{tgt} ”. This allows the model to align labels between source and target languages. We call this method **in-context label alignment**.

As opposed to in-context label alignment, we

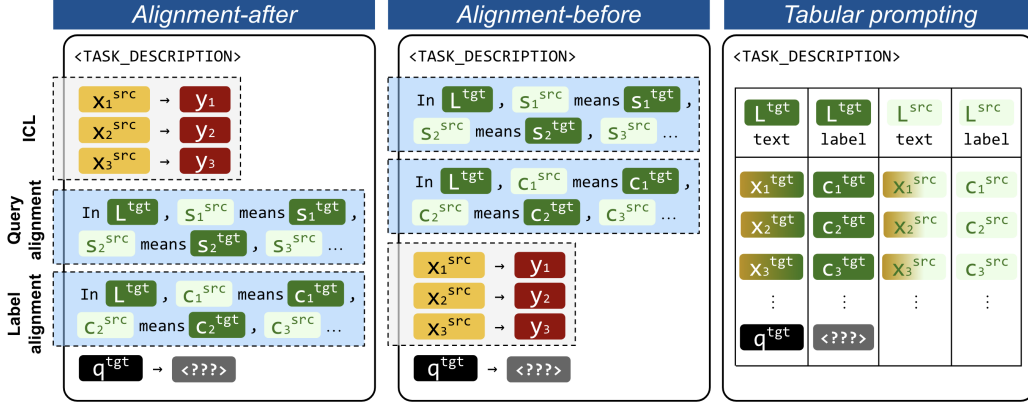


Figure 2: We explore 3 different alignment formats for X-ICL prompting, i.e., alignment-after, alignment-before, and tabular-prompting. From left to right, the prompt format has a higher degree of format consistency.

explore another approach, dubbed **in-context query alignment**, which provides alignment of input distribution by providing the translation of sentences similar to the query while keeping the label set as is. To do so, we utilize the parallel exemplar dataset $D^{para} = \{(s_1^{src}, s_1^{tgt}), (s_2^{src}, s_2^{tgt}), \dots, (s_m^{src}, s_m^{tgt})\}$, where (s_i^{src}, s_i^{tgt}) respectively denotes to a pair of parallel source and target sentences, and select the top-k most similar parallel pair by maximizing the monolingual similarity between the q^{tgt} with s_i^{tgt} . Given a target language L^{tgt} , the parallel pairs are then formatted into an input alignment prompt, i.e., “In L^{tgt} , s_1^{src} means s_1^{tgt} , s_2^{src} means s_2^{tgt} , ..., and s_k^{src} means s_k^{tgt} ”. We show the example query for in-context label alignment and in-context query alignment in Appendix B.

3.2 Cross-Lingual Prompting

Although cross-lingual in-context alignment has shown improvements as reported in (Tanwar et al., 2023), it introduces distortions to certain aspects of the Bayesian inference framework (Xie et al., 2022; Min et al., 2022) underlying in-context learning. Notably, this method compromises the formatting consistency and output distribution of the prompt. While the label aligner is expected to align the output distribution between the source and target labels, it is merely an idealistic assumption, which might not be the case in real cases. To better align with the Bayesian inference framework, we explore two cross-lingual prompt adjustments, i.e., alignment formatting and label configuration.

Alignment Formatting Existing X-ICL with alignment approach (Tanwar et al., 2023) places the alignment between ICL exemplars and the in-

put query (dubbed as **alignment-after**). We argue that such abrupt changes in the prompt format might cause performance degradation. To improve the format consistency, we also explore two prompt formats for X-ICL: 1) **alignment-before** and 2) **tabular-prompting**. Alignment-before simply swaps the alignment text with the cross-lingual exemplars. This avoids the abrupt format change between the exemplars and the query such that the neighboring text span is more format-consistent. Tabular-prompting formats the prompt in the form of a table with multiple columns, which allows a consistent prompt format, but at the same time requires either a labeled parallel corpus or disrupting the input-output mapping through incorrect labeling (Min et al., 2022). The depiction of the prompt formats and the X-ICL alignment is in Figure 2.

Label Configuration To improve the output distribution consistency, we explore alternatives of using either **source-only labels** and **target-only labels** as opposed to **in-context label alignment** which shifts the language of the labels from the source language in the exemplars to the target language in q^{tgt} . These alternatives serve as a comparison to measure the effectiveness of in-context label alignment. In this study, we focus on English as the source language. Appendix F analyzes the use of other closely related source languages.

3.3 Cross-Lingual Retrieval

Another way to improve X-ICL performance is by improving the exemplar retrieval quality. Given an input query q^{tgt} and a source language exemplar dataset $D^{src} = \{(e_1^{src}, y_1^{src}), (e_2^{src}, y_2^{src}), \dots, (e_n^{src}, y_n^{src})\}$, where e_1^{src} and y_1^{src} respectively denote the input and la-

Dataset	# Lang	# Unseen BLOOM	# Unseen XGLM	# Lang Family	Region(s)
NusaTranslation	6	6	6	1	Southeast Asia
MasakhaNews	9	4	8	3	Africa
AmericasNLI	10	10	9	8	South America
Tweet Sentiment Multilingual	7	2	0	2	Northern Africa, Europe, Central Asia

Table 1: The datasets and languages under study. Our study covers 25 low-resource languages and 7 relatively higher-resource languages from various regions.

bel of the exemplar, the goal of cross-lingual retrieval is to retrieve one or more labeled exemplars (e_i^{src}, y_i^{src}) semantically relevant to q^{tgt} . Most prior works in X-ICL (Winata et al., 2021a; Asai et al., 2023; Zhang et al., 2023; Lin et al., 2022a) incorporate random retrieval, while recently, Tanwar et al. (2023) utilize **cross-lingual semantic similarity** which significantly improves performance over the random retrieval.

Nevertheless, we argue that this approach might not be optimal in the case of low-resource languages as the semantic representation for these languages might not be well aligned with the high-resource languages (see Appendix C). Thus, we explore **translation semantic similarity** as an alternative. It performs monolingual semantic similarity on q^{tgt} to obtain a sentence in L^{tgt} from a parallel dataset D^{para} , then uses monolingual semantic similarity on its pair in L^{src} to find the high-resource exemplars from D^{src} . Although the monolingual semantic similarity between two sentences from a low-resource language is also sub-optimal, this problem can be alleviated by incorporating other similarity metrics such as TF-IDF and bag-of-words. We denote ICL method using the translation semantic similarity as **T-ICL**. We show this analysis in Appendix C along with the depiction of the cross-lingual retrieval methods.

4 Experimental Settings

4.1 Retrieval and In-Context Learning Setup

To calculate the cross-lingual and monolingual semantic similarity, we utilize multilingual sentence transformers (Reimers and Gurevych, 2019, 2020).² For all ICL experiments, we conduct ICL with 3-shot ICL exemplars. We run our experiments using two LLMs: XGLM-7.5B (Lin et al., 2022b) and BLOOM-7B (Scao et al., 2022). To select the prediction label, we take the label that

²As our semantic similarity model, we utilize sentence-transformers/stsb-xlm-r-multilingual

Eval Dataset	D^{src}	D^{para}
NusaTranslation	NusaX-Senti (Winata et al., 2022b)	NusaX-MT (Winata et al., 2022b)
MasakhaNews	MasakhaNews (Eng Train set)	MAFAND (Adelani et al., 2022a)
AmericasNLI	XNLI (Eng) (Conneau et al., 2018)	XNLI (Eng) $\oplus$ AmericasNLI (Dev set)*
Tweet Sentiment Multilingual	Tweet Sentiment Multilingual (Eng Train set)	Tweet Sentiment Multilingual (Eng MT) [†]

Table 2: The D^{src} and D^{para} for all the evaluation datasets under study.[†] Translated to English using NLLB (Team et al., 2022).* We align the two datasets.

maximizes the marginal probability of the prompt:

$$c^{pred} = \arg \max_c P(X^{icl}, X^{align}, q^{tgt}, c) \quad (1)$$

$$= f(X^{icl} \oplus X^{align} \oplus q^{tgt} \oplus c) \quad (2)$$

where $f(\cdot)$ denotes a language model, $\oplus$ denotes the concatenation operator, X^{icl} denotes the ICL exemplars, X^{align} denotes the alignment text, and c denotes the class label taken from the label set.

4.2 Languages and Datasets

As shown in Table 1, our study includes 25 low-resource languages from three different regions, i.e., Africa, Americas, and South-East Asia, covering 13 language families. Note that, many of the low-resource languages are unseen to both XGLM and BLOOM, nonetheless, both models might have seen other languages under the same language family group with those low-resource languages, e.g., both models are pre-trained on Indonesian, which falls under the same language family group (i.e., Malayo-Polynesian) to the low-resource languages in Indonesia. We also include 7 relatively higher-resource languages, i.e., Arabic (arb), French (fra), German (deu), Hindi (hin), Italian (ita), Portuguese (por), and Spanish (spa) for comparing the behavior of X-ICL between these relatively higher-resource languages and low-resource languages. Detailed information on all the languages under study is shown in Appendix A.

All the languages are spread across four different datasets, i.e., MasakhaNews (**topic classification**) (Adelani et al., 2023), AmericasNLI (**natural language inference**) (Ebrahimi et al., 2022b), NusaTranslation (**sentiment analysis**) (Cahyawijaya et al., 2023b), and TweetSentimentMultilingual (**sentiment analysis**) (Barbieri et al., 2022). For each dataset, we defined the ICL dataset D^{src} and parallel alignment dataset D^{para} from different dataset subsets or completely different datasets.

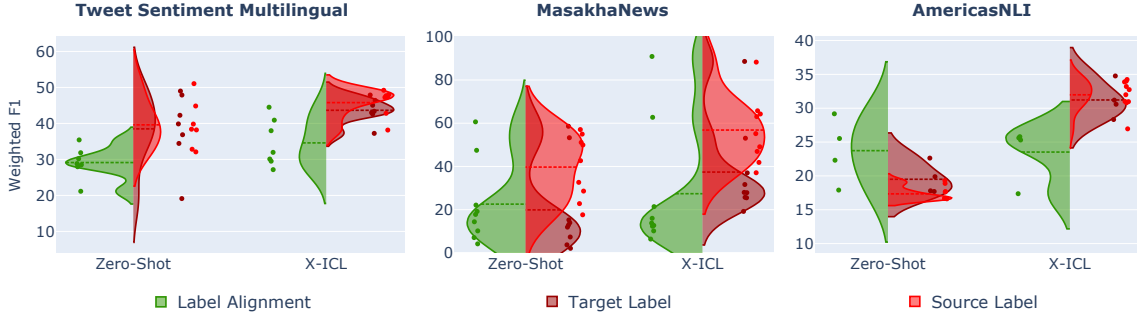


Figure 3: Performance of XGLM-7.5B with in-context label alignment, target-only label, and source-only label on (left) higher-resource, (center) low-resource African, and (right) low-resource American languages.

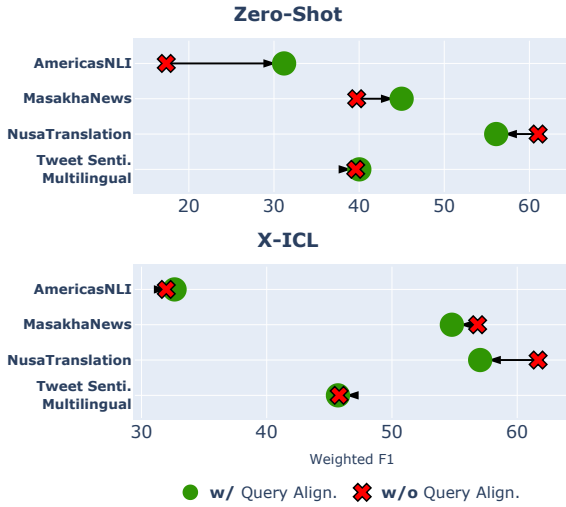


Figure 4: Performance of XGLM-7.5B with and without query alignment on (top) zero-shot and (bottom) X-ICL settings.

The details are shown in Table 2. Note that, we only take languages that are supported in NLLB (Team et al., 2022), such that we can compare the performance with machine-translation-based approaches. For the monolingual semantic similarity baselines, we utilize the train and dev sets of the evaluation dataset.

5 Result and Discussion

The per-dataset results of our experiments are shown in Appendix H. For brevity, we report the analysis mainly for XGLM-7.5B, since we observe that the BLOOM-7B results are similar. We show the results for BLOOM in Appendix G.

5.1 Inferiority of In-Context Label Alignment

Figure 3 shows the comparison of **in-context label alignment** with uniform **source-only** and **target-only** labels. In most languages, in-context label alignment yields lower performance than target-only label, and source-only label yields the best

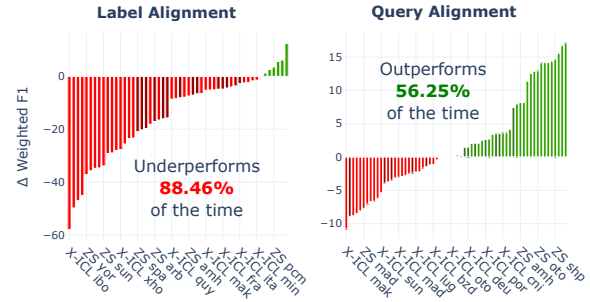


Figure 5: Δ Weighted F1 of (left) in-context label alignment and (right) in-context query alignment against non-alignment baseline. A score < 0 indicates the in-context alignment degrades the performance.

performance. For low-resource African languages, the target-only label performs much worse. We conjecture that this is due to the weak representation of these languages, which is less apparent in low-resource Indonesian and American languages because the target labels (see Appendix D) are similar to higher-resource languages in training. Contrary to Tanwar et al. (2023), our results highlight the ineffectiveness of **in-context label alignment** to improve X-ICL on both higher-resource and low-resource languages.

5.2 In-Context Query Alignment

We introduce in-context query alignment as an alternative to in-context label alignment in §3.1. As shown in Figure 4, **in-context query alignment** yields similar performance with the baseline (i.e., without query alignment) on higher-resource languages while improving zero-shot performance on low-resource languages. Nonetheless, the improvement is rather marginal in the X-ICL setting on low-resource languages. In this case, we conclude that in-context query alignment can be used as an alternative to X-ICL, which is favorable when there is no available X-ICL corpus for the particular task. With the recent development of large multilingual

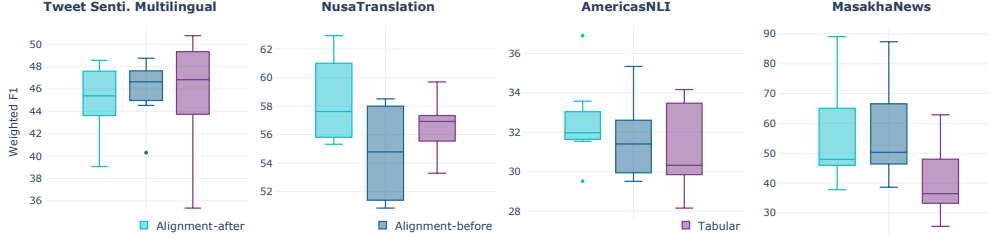


Figure 6: Performance of XGLM-7.5B with different alignment formats ordered by the degree of formatting consistency on (1) higher-resource languages, (2) low-resource Indonesian languages, (3) low-resource American languages, and (4) low-resource African languages.

parallel corpora, such as Bloom Library (Leong et al., 2022), WikiMatrix (Schwenk et al., 2021), CC-Aligned (Chaudhary et al., 2019; El-Kishky et al., 2020), FLORES-200 (Team et al., 2022), and GATITOS (Jones et al., 2023), in-context query alignment can also be a perfect complement to X-ICL for improving LLMs understanding on thousands of languages.

Label Alignment vs Query Alignment To investigate how well in-context alignments can affect the understanding of all the languages under study, we analyze their effectiveness by comparing them with the corresponding non-alignment baseline. As shown in Figure 5, in-context label alignment only improves the performance at $\sim 11.54\%$ of the time with an improvement of $\sim 5\%$ weighted F1, while the rest 88.46% experiments are decreased by $\sim 20\%$ weighted F1. In-context query alignment, on the other hand, increases the performance 56.25% of the time with an improvement of $\sim 10\%$ weighted F1, while the rest 43.75% of the time experiences a reduction of $\sim 5\%$ weighted F1. Our results suggest that **in-context query alignment** is superior to **in-context label alignment**, and it improves LLMs’ understanding of low-resource languages in the absence of X-ICL task-specific data, which leads to performance gain.

5.3 Why Query Alignment Performs Better

In regards to the in-context alignment, we can first simplify the effect of alignment into the following two possibilities: 1) when the alignment is successful (upper bound) and when no alignment is done (lower bound). When The LLM successfully aligns the query in the source language to the target language, query alignment will enable the LLM to understand the query in the target as well as in the source languages, the LLM will reach a performance similar to monolingual ICL, which is the upper bound performance. While in label alignment,

when the LLM successfully aligns the label in the source to the target languages, the LLM understands the label semantics, but there is no guidance on how to interpret the query in the target language. In this case, the upper bound is equivalent to performing X-ICL which generally performs slightly worse than monolingual high-resource language ICL which is reflected in our result in §5.6.

When the LLM completely fails to align the label in the source to the target languages, in the query alignment, the LLM performs a regular X-ICL, which is similar to the best case of the label alignment. While in the label alignment, the LLM performs X-ICL with a shifted label space. The harmful effect of ICL with a shifted label space has been extensively studied in (Min et al., 2022), which results in severe performance degradation. With regards to the two possibilities, the upper-bound and lower-bound of in-context query alignment are better than label alignment, thus query alignment outperforms label alignment on average. In a more realistic scenario, there is also another factor where the alignment text becomes the noise that will shift the output prediction of the LLMs. As the noise factor happens for both query and label alignment, we can assume the same effect of noise for both methods and omit this factor into account, resulting in the same conclusion.

5.4 Effect of Format Consistency

We explore three types of prompting with various degrees of formatting consistency (§3.2). As shown in Figure 6, for higher-resource languages, formatting consistency correlates to a slight improvement in the downstream performance for both XGLM and BLOOM (see Appendix G) models. Meanwhile, for low-resource languages, the trend for both models is unclear. We conjecture that increasing the format consistency can improve the downstream task performance on well-represented

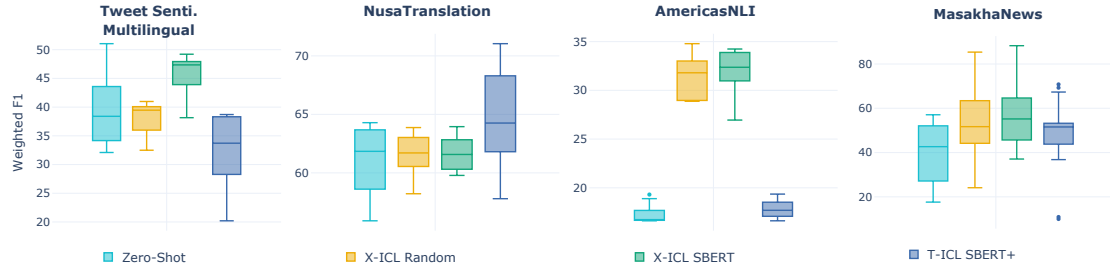


Figure 7: Performance of XGLM-7.5B with different in-context learning retrievals covering monolingual, cross-lingual, translation semantic similarity (T-ICL) on (1) higher-resource languages, (2) low-resource Indonesian languages, (3) low-resource American languages, and (4) low-resource African languages. Random and SBERT denotes random and semantic-similarity-based exemplar selection, respectively.

languages. For low-resource languages, increasing the format consistency will not improve the model understanding. Increasing the representation through X-ICL and query alignment would be a better alternative to improve the low-resource language understanding ability of LLMs.

5.5 Importance of Cross-Lingual Retrieval

Cross-Lingual Semantic Similarity We compare the effectiveness of cross-lingual semantic similarity to monolingual and translation semantic similarity for retrieving ICL and X-ICL exemplars. Based on Figure 7, X-ICL and ICL with cross-lingual and monolingual semantic-similarity-based retrieval, respectively, perform better than zero-shot prompting, suggesting the effectiveness of these approaches for improving the task understanding of LLMs. In addition, we show that translation semantic similarity performs almost on par with the zero-shot baseline. We hypothesize that this problem is attributed to the error propagation of the pipelined nature of the translation semantic similarity system and the limited coverage of parallel exemplars in D^{para} , showing the benefit of using direct cross-lingual semantic similarity retrieval over translation-based retrieval. Furthermore, the performance of cross-lingual semantic similarity is similar to or slightly lower than the monolingual semantic similarity approach. Hence, cross-lingual semantic similarity retrieval is important in the case where the corpus for performing monolingual ICL on a particular task is not available.

Variations of Semantic Similarity Models We further compare the effectiveness of varying the cross-lingual semantic similarity models for cross-lingual retrieval. As shown in Figure 8, all cross-lingual semantic similarity models outperform the zero-shot baseline. Interestingly, despite

the reported inferiority of STS-tuned models over paraphrasing-tuned models and LaBSE in prior works (Reimers and Gurevych, 2019, 2020; Feng et al., 2022), our results showcase otherwise. On average, XLMR STS performs on par with other models, gaining a better performance on high-resource languages while getting a worse performance on low-resource languages. We find that, depending on the language under study, the choice of cross-lingual semantic similarity models can play a huge role in the downstream performance of X-ICL.

5.6 Is X-ICL Effective for low-resource Languages?

To analyze the effectiveness of X-ICL in low-resource languages, we compare X-ICL with other inference approaches. Specifically, we compare X-ICL with 3 other baselines: 1) monolingual **ICL** that performs inference using ICL from the same language as the query, 2) **translate-test** that translates the query and performs zero-shot inference in a high-resource language, i.e., English, and 3) **translate-test ICL** that simply combines **translate-test** and monolingual **ICL**. We measure the Δ Weighted F1 against a simple **zero-shot prompting** over all languages under study. For all experiments that include translation, we utilize MT models from NLLB (Team et al., 2022).³

Based on our experiment results shown in Figure 9, the **translate-test** slightly improves the performance from the zero-shot baseline in BLOOM and XGLM, while **in-context query alignment** only improves zero-shot performance on XGLM. This indicates that alignment information only offers a limited benefit to improving LLMs’ understanding. Additionally, all ICL approaches improve the performance over zero-shot prompting in most

³<https://huggingface.co/facebook/nllb-200-distilled-1.3B>

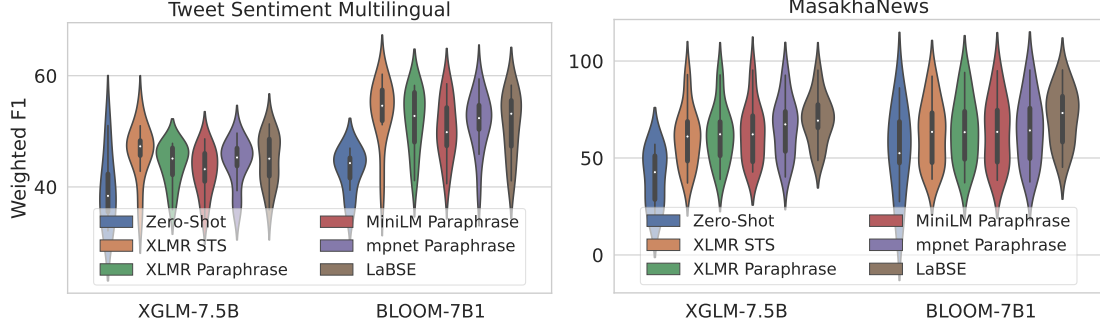


Figure 8: Performance of LLMs with different semantic similarity models on **(left)** higher-resource languages and **(right)** low-resource African languages.

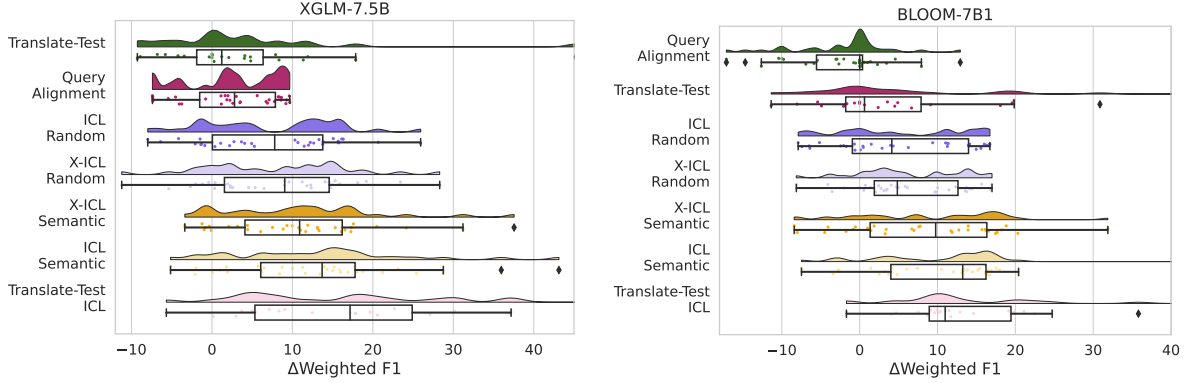


Figure 9: Gain/Loss of various test-time adaptation methods for low-resource languages using **(top)** XGLM-7.5B and **(bottom)** BLOOM-7B1 backbones.

cases. All approaches with similarity-based retrieval, i.e., **ICL Semantic** and **X-ICL Semantic** achieve higher scores than random retrievals, i.e., **ICL Random** and **X-ICL Random**, showing the importance of semantic similarity for exemplar retrievals. Interestingly, **X-ICL Semantic** yields a similar performance to **ICL Semantic**, which utilizes the target language exemplars. This indicates X-ICL can be a good alternative for low-resource languages as the available data in the specified low-resource language are commonly very limited. Above all, **translate-test ICL** yields the highest improvement amongst all methods, but this only happens when the machine translation quality is above a certain quality standard. We ablate the effect of machine translation quality to the **translate-test ICL** performance on Appendix E.

To conclude, we offer the following suggestions to improve the low-resource language performance during inference: 1) When tackling low-resource languages, it is best to have a high-quality translation system accompanied by a source language task-specific data for **translate-test ICL**; 2) When there is no machine translation (MT) system for the specified language, it is best to use either **ICL** or **X-ICL** depending on the corpus availability; 3)

When there is an MT system, but no task-specific data, **translate-test** is still the best option; and 4) When there is no high-quality MT system nor task-specific data, the best way is to use a parallel data to utilize **in-context query-alignment**.

6 Conclusion

We systematically investigate the application of X-ICL with LLMs, focusing on low-resource languages. Our comprehensive analysis sheds light on multiple facets of X-ICL with LLMs. Our examination of in-context alignment reveals the limitation of label alignment, thus we suggest a more effective alternative: query alignment. Efforts to enhance X-ICL via formatting consistency only exhibit a marginal impact on low-resource languages. Our exploration of exemplar retrieval approaches underscores the significance of employing cross-lingual semantic similarity in X-ICL. Lastly, we analyze the effectiveness of X-ICL in the context of low-resource languages. Despite being outperformed by translate-test ICL, X-ICL remains relevant, especially when there is no MT model available for the target language—a circumstance prevalent in low-resource language scenarios.

Acknowledgements

This work is partially funded by the PF20-43679 Hong Kong PhD Fellowship Scheme, Research Grant Council, Hong Kong; the Hong Kong Fellowship Scheme by the Hong Kong Research Grants Council (RGC); and the National Research Foundation, Singapore under its AI Singapore Programme.

Ethics Statement

Our exploration of ICL and X-ICL methods addresses the linguistic data gap in low-resource languages. In scenarios where no monolingual corpus or machine translation system exists, our work underscores the significance of ICL and X-ICL as a viable solution. By investigating the limitations of in-context label alignment and proposing a more effective in-context query alignment approach, we aim to enhance the applicability of ICL and X-ICL on low-resource languages. This research is motivated by the need to provide computational solutions for languages lacking adequate linguistic resources for LM training. Our findings emphasize that ICL and X-ICL are useful in scenarios where alternative resources are absent, promoting linguistic diversity and inclusivity in the development of language technologies. All the datasets used in our experiments follow the license and term of use of the datasets.

Limitation

Limited Coverage of low-resource Languages

We put our best effort into collecting datasets from various low-resource languages and, in the end, we ended up with the three low-resource datasets, i.e., MasakhaNews (Adelani et al., 2023), NusaTranslation (Cahyawijaya et al., 2023b), and AmericasNLI (Ebrahimi et al., 2022a), which suits our cases as these languages have parallel datasets which correspond to one or more high-resource languages and have large enough high-quality labeled datasets for both ICL and evaluation purposes. Furthermore, our study covers broad enough linguistics aspects of multilingual and cross-lingual within these three datasets, including various linguistics distances with the source languages — from Nigerian Pidgin (pcm) to obscure regional languages such as Batak (btw), Hausa (hau), and Guarani (grn) —, broad linguistic and geographic diversity —the low-resource languages under study covers >10 language families from three different continents —, and the incorporation of different scripts

between source and target languages — in the case of Amharic as a low-resource language and Arabic as a higher-resource language —. We leave the study of other and broader scales of low-resource languages for future work.

Choice of Multilingual High-Resource Language Datasets

Prior work on X-ICL with alignment (Tanwar et al., 2023) conduct their study on Multilingual Amazon Review Corpus (MARC) (Keung et al., 2020), Cross-language Sentiment (CLS) (Prettenhofer and Stein, 2010), and HatEval (Basile et al., 2019). We considered using these datasets as our high-resource languages dataset. Nonetheless, we found that both MARC and CLS datasets are no longer available⁴, leaving us with only HatEval dataset. Since HatEval only covers English and Spanish, we do not incorporate it in our study. Instead, we incorporate the TweetSentimentMultilingual dataset (Barbieri et al., 2022) which covers 7 relatively high-resource languages in our study. We leave the exploration of other high-resource languages to future work.

Task Coverage Given the nature of low-resource languages, there are only a handful of datasets available as downstream tasks. We suggest future works to explore the generalization of our approach to a broader task coverage, especially on datasets that cover more culturally relevant nuances of the corresponding low-resource language (Aji et al., 2022b; Kabra et al., 2023; Lovenia et al., 2024).

Exploration on Other LLMs We conduct our experiments with a single RTX3090 (24GB) GPU. Due to the large cost of inference and limited computation budget, we do not experiment on larger multilingual LLMs such as Falcon (Almazrouei et al., 2023) and MPT (Team, 2023). Nonetheless, exploration on incorporating ICL and X-ICL with random exemplar retrieval with larger LLMs and the scaling effect on low-resource settings, such as low-resource languages and code-switching, have been discussed in prior works (Zhang et al., 2023; Asai et al., 2023; Cahyawijaya et al., 2024). We hypothesize that the scaling behavior of our work will follow the same trend.

⁴We checked the MARC dataset from the Hugging Face hub URL (https://huggingface.co/datasets/amazon_reviews_multi) and the original Amazon Web Service S3 Bucket (<https://s3.console.aws.amazon.com/s3/buckets/amazon-reviews-m1>). While for the CLS dataset, we checked the original dataset link in the paper (<http://www.webis.de/research/corpora/webis-cls-10/>).

References

- David Adelani, Jesujoba Alabi, Angela Fan, Julia Kreutzer, Xiaoyu Shen, Machel Reid, Dana Ruitter, Dietrich Klakow, Peter Nabende, Ernie Chang, Tajuddeen Gwadabe, Freshia Sackey, Bonaventure F. P. Dossou, Chris Emezue, Colin Leong, Michael Beukman, Shamsuddeen Muhammad, Guyo Jarso, Oreen Yousuf, Andre Niyongabo Rubungo, Gilles Hacheme, Eric Peter Wairagala, Muhammad Umair Nasir, Benjamin Ajibade, Tunde Ajayi, Yvonne Gitau, Jade Abbott, Mohamed Ahmed, Millicent Ochieng, Anuoluwapo Aremu, Perez Ogayo, Jonathan Mukiibi, Fatoumata Ouoba Kabore, Godson Kalipe, Derguene Mbaye, Allahsera Auguste Tapo, Victoire Memdjokam Koagne, Edwin Munkoh-Buabeng, Valencia Wagner, Idris Abdulmumin, Ayodele Awokoya, Happy Buzaaba, Blessing Sibanda, Andiswa Bukula, and Sam Manthulu. 2022a. [A few thousand translations go a long way! leveraging pre-trained models for African news translation](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3053–3070, Seattle, United States. Association for Computational Linguistics.
- David Adelani, Graham Neubig, Sebastian Ruder, Shruti Rijhwani, Michael Beukman, Chester Palen-Michel, Constantine Lignos, Jesujoba Alabi, Shamsuddeen Muhammad, Peter Nabende, Cheikh M. Bamba Dione, Andiswa Bukula, Rooweither Mabuya, Bonaventure F. P. Dossou, Blessing Sibanda, Happy Buzaaba, Jonathan Mukiibi, Godson Kalipe, Derguene Mbaye, Amelia Taylor, Fatoumata Kabore, Chris Chinenye Emezue, Anuoluwapo Aremu, Perez Ogayo, Catherine Gitau, Edwin Munkoh-Buabeng, Victoire Memdjokam Koagne, Allahsera Auguste Tapo, Tebogo Macucwa, Vukosi Mavate, Mboning Tchiase Elvis, Tajuddeen Gwadabe, Tosin Adewumi, Orevaoghene Ahia, Joyce Nakatumba-Nabende, Neo Lerato Mokono, Ignatius Ezeani, Chiamaka Chukwuneke, Mofetoluwa Oluwaseun Adeyemi, Gilles Quentin Hacheme, Idris Abdulmumin, Odunayo Ogundepo, Oreen Yousuf, Tatiana Moteu, and Dietrich Klakow. 2022b. [MasakhaNER 2.0: Africa-centric transfer learning for named entity recognition](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 4488–4508, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- David Ifeoluwa Adelani, Jade Abbott, Graham Neubig, Daniel D'souza, Julia Kreutzer, Constantine Lignos, Chester Palen-Michel, Happy Buzaaba, Shruti Rijhwani, Sebastian Ruder, Stephen Mayhew, Israel Abebe Azime, Shamsuddeen H. Muhammad, Chris Chinenye Emezue, Joyce Nakatumba-Nabende, Perez Ogayo, Aremu Anuoluwapo, Catherine Gitau, Derguene Mbaye, Jesujoba Alabi, Seid Muhie Yimam, Tajuddeen Rabi'u Gwadabe, Ignatius Ezeani, Rubungo Andre Niyongabo, Jonathan Mukiibi, Verah Otiende, Iroko Orife, Davis David, Samba Ngom, Tosin Adewumi, Paul Rayson, Mofetoluwa Adeyemi, Gerald Muriuki, Emmanuel Anebi, Chiamaka Chukwuneke, Nkiruka Odu, Eric Peter Wairagala, Samuel Oyerinde, Clemencia Siro, Tobius Saul Bateesa, Temilola Oloyede, Yvonne Wambui, Victor Akinode, Deborah Nabagereka, Maurice Katusiime, Ayodele Awokoya, Mouhamadane MBOUP, Dibora Gebreyohannes, Henok Tilaye, Kelechi Nwaike, Degaga Wolde, Abdoulaye Faye, Blessing Sibanda, Orevaoghene Ahia, Bonaventure F. P. Dossou, Kelechi Ogueji, Thierno Ibrahima DIOP, Abdoulaye Diallo, Adewale Akinfaderin, Tendai Marengereke, and Salomey Osei. 2021. [MasakhaNER: Named Entity Recognition for African Languages](#). *Transactions of the Association for Computational Linguistics*, 9:1116–1131.
- David Ifeoluwa Adelani, Marek Masiak, Israel Abebe Azime, Jesujoba Alabi, Atnafu Lambebo Tonja, Christine Mwase, Odunayo Ogundepo, Bonaventure F. P. Dossou, Akintunde Oladipo, Doreen Nixdorf, Chris Chinenye Emezue, sana al azzawi, Blessing Sibanda, Davis David, Lolwethu Ndoela, Jonathan Mukiibi, Tunde Ajayi, Tatiana Moteu, Brian Odhiambo, Abraham Owodunni, Nnaemeka Obiefuna, Muhidin Mohamed, Shamsuddeen Hassan Muhammad, Teshome Mulugeta Ababu, Saheed Abdulahi Salahudeen, Mesay Gemeda Yigezu, Tajuddeen Gwadabe, Idris Abdulmumin, Mahlet Taye, Oluwabusayo Awoyomi, Iyanuoluwa Shode, Tolulope Adelani, Habiba Abdulganiyu, Abdul-Hakeem Omatayo, Adetola Adeeko, Abeeb Afolabi, Anuoluwapo Aremu, Olanrewaju Samuel, Clemencia Siro, Wangari Kimotho, Onyekachi Ogbu, Chinedu Mbonu, Chiamaka Chukwuneke, Samuel Fanijo, Jessica Ojo, Oyinkansola Awosan, Tadesse Kebede, Toadum Sari Sakayo, Pamela Nyatsine, Freedom Sidume, Oreen Yousuf, Mardiyyah Odwole, Tshinu Tshinu, Ussen Kimanuka, Thina Diko, Siyanda Nxakama, Sinodos Nigusse, Abdulmejid Johar, Shafie Mohamed, Fuad Mire Hassan, Moges Ahmed Mehamed, Evrard Ngabire, Jules Jules, Ivan Senkungu, and Pontus Stenetorp. 2023. [Masakhanews: News topic classification for african languages](#).
- Muhammad Farid Adilazuarda, Samuel Cahyawijaya, Alham Fikri Aji, Genta Indra Winata, and Ayu Purwarianti. 2024. [Lingualchemy: Fusing typological and geographical elements for unseen language generalization](#).
- Muhammad Farid Adilazuarda, Samuel Cahyawijaya, and Ayu Purwarianti. 2023. [The obscure limitation of modular multilingual language models](#).
- Alham Fikri Aji, Genta Indra Winata, Fajri Koto, Samuel Cahyawijaya, Ade Romadhony, Rahmad Mahendra, Kemal Kurniawan, David Moeljadi, Radityo Eko Prasoj, Timothy Baldwin, Jey Han Lau, and Sebastian Ruder. 2022a. [One country, 700+ languages: NLP challenges for underrepresented languages and dialects in Indonesia](#). In *Proceedings of the 60th Annual Meeting of the Association for*

- Computational Linguistics (Volume 1: Long Papers)*, pages 7226–7249, Dublin, Ireland. Association for Computational Linguistics.
- Alham Fikri Aji, Genta Indra Winata, Fajri Koto, Samuel Cahyawijaya, Ade Romadhony, Rahmad Mahendra, Kemal Kurniawan, David Moeljadi, Radityo Eko Prasajo, Timothy Baldwin, Jey Han Lau, and Sebastian Ruder. 2022b. [One country, 700+ languages: NLP challenges for underrepresented languages and dialects in Indonesia](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7226–7249, Dublin, Ireland. Association for Computational Linguistics.
- Ebtesam Almazrouei, Hamza Alobeidli, Abdulaziz Alshamsi, Alessandro Cappelli, Ruxandra Cojocaru, Merouane Debbah, Etienne Goffinet, Daniel Hestlow, Julien Launay, Quentin Malartic, Badreddine Noune, Baptiste Pannier, and Guilherme Penedo. 2023. Falcon-40B: an open large language model with state-of-the-art performance.
- Akari Asai, Sneha Kudugunta, Xinyan Velocity Yu, Terra Blevins, Hila Gonen, Machel Reid, Yulia Tsvetkov, Sebastian Ruder, and Hannaneh Hajishirzi. 2023. [Buffet: Benchmarking large language models for few-shot cross-lingual transfer](#).
- Lucas Bandarkar, Davis Liang, Benjamin Muller, Mikel Artetxe, Satya Narayan Shukla, Donald Husa, Naman Goyal, Abhinandan Krishnan, Luke Zettlemoyer, and Madian Khabisa. 2023. The belebele benchmark: a parallel reading comprehension dataset in 122 language variants. *arXiv preprint arXiv:2308.16884*.
- Yejin Bang, Samuel Cahyawijaya, Nayeon Lee, Wenliang Dai, Dan Su, Bryan Wilie, Holy Lovenia, Ziwei Ji, Tiezheng Yu, Willy Chung, Quyet V. Do, Yan Xu, and Pascale Fung. 2023. [A multitask, multilingual, multimodal evaluation of chatgpt on reasoning, hallucination, and interactivity](#).
- Francesco Barbieri, Luis Espinosa Anke, and Jose Camacho-Collados. 2022. [XLM-T: Multilingual language models in Twitter for sentiment analysis and beyond](#). In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 258–266, Marseille, France. European Language Resources Association.
- Valerio Basile, Cristina Bosco, Elisabetta Fersini, Debora Nozza, Viviana Patti, Francisco Manuel Rangel Pardo, Paolo Rosso, and Manuela Sanguinetti. 2019. [SemEval-2019 task 5: Multilingual detection of hate speech against immigrants and women in Twitter](#). In *Proceedings of the 13th International Workshop on Semantic Evaluation*, pages 54–63, Minneapolis, Minnesota, USA. Association for Computational Linguistics.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Matiusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020a. [Language models are few-shot learners](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020b. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.
- Samuel Cahyawijaya, Holy Lovenia, Alham Fikri Aji, Genta Indra Winata, Bryan Wilie, Rahmad Mahendra, Christian Wibisono, Ade Romadhony, Karissa Vincentio, Fajri Koto, Jennifer Santoso, David Moeljadi, Cahya Wirawan, Frederikus Hudi, Ivan Halim Parmonangan, Ika Alfina, Muhammad Satrio Wicaksono, Ilham Firdausi Putra, Samsul Rahmadani, Yulianti Oenang, Ali Akbar Septiandri, James Jaya, Kaustubh D. Dhole, Arie Ardiyanti Suryani, Rifki Afina Putri, Dan Su, Keith Stevens, Made Nindyatama Nityasya, Muhammad Farid Adilazuarda, Ryan Ignatius, Ryandito Diandaru, Tiezheng Yu, Vito Ghifari, Wenliang Dai, Yan Xu, Dyah Damapusita, Cuk Tho, Ichwanul Muslim Karo Karo, Tirana Noor Fatyanosa, Ziwei Ji, Pascale Fung, Graham Neubig, Timothy Baldwin, Sebastian Ruder, Herry Sujaini, Sakriani Sakti, and Ayu Purwarianti. 2023a. [Nusacrowd: Open source initiative for indonesian nlp resources](#).
- Samuel Cahyawijaya, Holy Lovenia, Fajri Koto, Dea Adhista, Emmanuel Dave, Sarah Oktavianti, Salsabil Akbar, Jhonson Lee, Nur Shadieq, Tjeng Wawan Cenggoro, hanung linuwih, Bryan Wilie, Galih Muri-dan, Genta Winata, David Moeljadi, Alham Fikri Aji, Ayu Purwarianti, and Pascale Fung. 2023b. [Nusawrites: Constructing high-quality corpora for underrepresented and extremely low-resource languages](#).
- Samuel Cahyawijaya, Holy Lovenia, Fajri Koto, Rifki Afina Putri, Emmanuel Dave, Jhonson Lee, Nur Shadieq, Wawan Cenggoro, Salsabil Maulana Akbar, Muhammad Ihza Mahendra, Dea Annisayanti Putri, Bryan Wilie, Genta Indra Winata, Alham Fikri Aji, Ayu Purwarianti, and Pascale Fung. 2024. [Cendol: Open instruction-tuned generative large language models for indonesian languages](#).
- Samuel Cahyawijaya, Holy Lovenia, Tiezheng Yu, Willy Chung, and Pascale Fung. 2023c. [Instruct-align: Teaching novel languages with to llms through alignment-based cross-lingual instruction](#).
- Samuel Cahyawijaya, Genta Indra Winata, Bryan Wilie, Karissa Vincentio, Xiaohong Li, Adhiguna Kuncoro, Sebastian Ruder, Zhi Yuan Lim, Syafri Bahar, Masayu Khodra, Ayu Purwarianti, and Pascale

- Fung. 2021. [IndoNLG: Benchmark and resources for evaluating Indonesian natural language generation](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 8875–8898, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Vishrav Chaudhary, Yuqing Tang, Francisco Guzmán, Holger Schwenk, and Philipp Koehn. 2019. [Low-resource corpus filtering using multilingual sentence embeddings](#). In *Proceedings of the Fourth Conference on Machine Translation (Volume 3: Shared Task Papers, Day 2)*, pages 261–266, Florence, Italy. Association for Computational Linguistics.
- Arslan Chaudhry, Marcus Rohrbach, Mohamed Elhoseiny, Thalaiyasingam Ajanthan, Puneet K. Dokania, Philip H. S. Torr, and Marc’Aurelio Ranzato. 2019. [On tiny episodic memories in continual learning](#).
- Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, Parker Schuh, Kensen Shi, Sasha Tsvyashchenko, Joshua Maynez, Abhishek Rao, Parker Barnes, Yi Tay, Noam Shazeer, Vinodkumar Prabhakaran, Emily Reif, Nan Du, Ben Hutchinson, Reiner Pope, James Bradbury, Jacob Austin, Michael Isard, Guy Gur-Ari, Pengcheng Yin, Toju Duke, Anselm Levskaya, Sanjay Ghemawat, Sunipa Dev, Henryk Michalewski, Xavier Garcia, Vedant Misra, Kevin Robinson, Liam Fedus, Denny Zhou, Daphne Ippolito, David Luan, Hyeontaek Lim, Barret Zoph, Alexander Spiridonov, Ryan Sepassi, David Dohan, Shivani Agrawal, Mark Omernick, Andrew M. Dai, Thanumalayan Sankaranarayanan Pillai, Marie Pellat, Aitor Lewkowycz, Erica Moreira, Rewon Child, Oleksandr Polozov, Katherine Lee, Zongwei Zhou, Xuezhi Wang, Brennan Saeta, Mark Diaz, Orhan Firat, Michele Catasta, Jason Wei, Kathy Meier-Hellstern, Douglas Eck, Jeff Dean, Slav Petrov, and Noah Fiedel. 2022. [Palm: Scaling language modeling with pathways](#).
- Alexis Conneau, Ruty Rinott, Guillaume Lample, Adina Williams, Samuel Bowman, Holger Schwenk, and Veselin Stoyanov. 2018. [XNLI: Evaluating cross-lingual sentence representations](#). In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2475–2485, Brussels, Belgium. Association for Computational Linguistics.
- Abteen Ebrahimi, Manuel Mager, Arturo Oncevay, Vishrav Chaudhary, Luis Chiruzzo, Angela Fan, John Ortega, Ricardo Ramos, Annette Rios, Ivan Vladimir Meza Ruiz, Gustavo Giménez-Lugo, Elisabeth Mager, Graham Neubig, Alexis Palmer, Rolando Coto-Solano, Thang Vu, and Katharina Kann. 2022a. [AmericasNLI: Evaluating zero-shot natural language understanding of pretrained multilingual models in truly low-resource languages](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6279–6299, Dublin, Ireland. Association for Computational Linguistics.
- Abteen Ebrahimi, Manuel Mager, Arturo Oncevay, Vishrav Chaudhary, Luis Chiruzzo, Angela Fan, John Ortega, Ricardo Ramos, Annette Rios, Ivan Vladimir Meza Ruiz, Gustavo Giménez-Lugo, Elisabeth Mager, Graham Neubig, Alexis Palmer, Rolando Coto-Solano, Thang Vu, and Katharina Kann. 2022b. [AmericasNLI: Evaluating zero-shot natural language understanding of pretrained multilingual models in truly low-resource languages](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6279–6299, Dublin, Ireland. Association for Computational Linguistics.
- Ahmed El-Kishky, Vishrav Chaudhary, Francisco Guzmán, and Philipp Koehn. 2020. [CCAligned: A massive collection of cross-lingual web-document pairs](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5960–5969, Online. Association for Computational Linguistics.
- Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Ariavazhagan, and Wei Wang. 2022. [Language-agnostic BERT sentence embedding](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 878–891, Dublin, Ireland. Association for Computational Linguistics.
- Robert M. French. 1993. Catastrophic interference in connectionist networks: Can it be predicted, can it be prevented? In *Proceedings of the 6th International Conference on Neural Information Processing Systems, NIPS’93*, page 1176–1177, San Francisco, CA, USA. Morgan Kaufmann Publishers Inc.
- Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc’Aurelio Ranzato, Francisco Guzmán, and Angela Fan. 2022. [The Flores-101 evaluation benchmark for low-resource and multilingual machine translation](#). *Transactions of the Association for Computational Linguistics*, 10:522–538.
- Xisen Jin, Xiang Ren, Daniel Preotiuc-Pietro, and Pengxiang Cheng. 2023. [Dataless knowledge fusion by merging weights of language models](#). In *The Eleventh International Conference on Learning Representations*.
- Alex Jones, Isaac Caswell, Ishank Saxena, and Orhan Firat. 2023. [Bilex rx: Lexical data augmentation for massively multilingual machine translation](#).
- Anubha Kabra, Emmy Liu, Simran Khanuja, Alham Fikri Aji, Genta Winata, Samuel Cahyawijaya, Anuoluwapo Aremu, Perez Ogayo, and Graham Neubig. 2023. [Multi-lingual and multi-cultural figurative language understanding](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8269–8284, Toronto, Canada. Association for Computational Linguistics.

- Divyanshu Kakwani, Anoop Kunchukuttan, Satish Golla, Gokul N.C., Avik Bhattacharyya, Mitesh M. Khapra, and Pratyush Kumar. 2020. [IndicNLP Suite: Monolingual corpora, evaluation benchmarks and pre-trained multilingual language models for Indian languages](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4948–4961, Online. Association for Computational Linguistics.
- Phillip Keung, Yichao Lu, György Szarvas, and Noah A. Smith. 2020. [The multilingual Amazon reviews corpus](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4563–4568, Online. Association for Computational Linguistics.
- Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. [Large language models are zero-shot reasoners](#). In *ICML 2022 Workshop on Knowledge Retrieval and Language Models*.
- Fajri Koto, Nurul Aisyah, Haonan Li, and Timothy Baldwin. 2023. [Large language models only pass primary school exams in Indonesia: A comprehensive test on IndoMMLU](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12359–12374, Singapore. Association for Computational Linguistics.
- Aman Kumar, Himani Shrotriya, Prachi Sahu, Amogh Mishra, Raj Dabre, Ratish Puduppully, Anoop Kunchukuttan, Mitesh M. Khapra, and Pratyush Kumar. 2022. [IndicNLG benchmark: Multilingual datasets for diverse NLG tasks in Indic languages](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 5363–5394, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Colin Leong, Joshua Nemecek, Jacob Mansdorfer, Anna Filighera, Abraham Owodunni, and Daniel Whitenack. 2022. [Bloom library: Multimodal datasets in 300+ languages for a variety of downstream tasks](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 8608–8621, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Alexander Cosgrove, Christopher D Manning, Christopher Re, Diana Acosta-Navas, Drew Arad Hudson, Eric Zelikman, Esin Durmus, Faisal Ladhak, Frieda Rong, Hongyu Ren, Huaxiu Yao, Jue WANG, Keshav Santhanam, Laurel Orr, Lucia Zheng, Mert Yuksekgonul, Mirac Suzgun, Nathan Kim, Neel Guha, Niladri S. Chatterji, Omar Khattab, Peter Henderson, Qian Huang, Ryan Andrew Chi, Sang Michael Xie, Shibani Santurkar, Surya Ganguli, Tatsunori Hashimoto, Thomas Icard, Tianyi Zhang, Vishrav Chaudhary, William Wang, Xuechen Li, Yifan Mai, Yuhui Zhang, and Yuta Ko-reeda. 2023. [Holistic evaluation of language models](#). *Transactions on Machine Learning Research*. Featured Certification, Expert Certification.
- Xi Victoria Lin, Todor Mihaylov, Mikel Artetxe, Tianlu Wang, Shuohui Chen, Daniel Simig, Myle Ott, Naman Goyal, Shruti Bhosale, Jingfei Du, Ramakanth Pasunuru, Sam Shleifer, Punit Singh Koura, Vishrav Chaudhary, Brian O’Horo, Jeff Wang, Luke Zettlemoyer, Zornitsa Kozareva, Mona Diab, Veselin Stoyanov, and Xian Li. 2022a. [Few-shot learning with multilingual generative language models](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 9019–9052, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Xi Victoria Lin, Todor Mihaylov, Mikel Artetxe, Tianlu Wang, Shuohui Chen, Daniel Simig, Myle Ott, Naman Goyal, Shruti Bhosale, Jingfei Du, Ramakanth Pasunuru, Sam Shleifer, Punit Singh Koura, Vishrav Chaudhary, Brian O’Horo, Jeff Wang, Luke Zettlemoyer, Zornitsa Kozareva, Mona Diab, Veselin Stoyanov, and Xian Li. 2022b. [Few-shot learning with multilingual language models](#).
- Holy Lovenia, Wenliang Dai, Samuel Cahyawijaya, Ziwei Ji, and Pascale Fung. 2023. Negative object presence evaluation (nope) to measure object hallucination in vision-language models. *arXiv preprint arXiv:2310.05338*.
- Holy Lovenia, Rahmad Mahendra, Salsabil Maulana Akbar, Lester James V. Miranda, Jennifer Santoso, Elyanah Aco, Akhdan Fadhillah, Jonibek Mansurov, Joseph Marvin Imperial, Onno P. Kampman, Joel Ruben Antony Moniz, Muhammad Ravi Shulthan Habibi, Frederikus Hudi, Railey Montalan, Ryan Ignatius, Joanito Agili Lopo, William Nixon, Börje F. Karlsson, James Jaya, Ryandito Diandaru, Yuze Gao, Patrick Amadeus, Bin Wang, Jan Christian Blaise Cruz, Chenxi Whitehouse, Ivan Halim Parmonangan, Maria Khelli, Wenyu Zhang, Lucky Susanto, Reynard Adha Ryanda, Sonny Lazuardi Hermawan, Dan John Velasco, Muhammad Dehan Al Kautsar, Willy Fitra Hendria, Yasmin Moslem, Noah Flynn, Muhammad Farid Adilazuarda, Haochen Li, Johanes Lee, R. Damanhuri, Shuo Sun, Muhammad Reza Qorib, Amirbek Djanibekov, Wei Qi Leong, Quyet V. Do, Niklas Muennighoff, Tanrada Pansuwan, Ilham Firdausi Putra, Yan Xu, Ngee Chia Tai, Ayu Purwarianti, Sebastian Ruder, William Tjhi, Peerat Limkonchotiwat, Alham Fikri Aji, Sedrick Keh, Genta Indra Winata, Ruochen Zhang, Fajri Koto, Zheng-Xin Yong, and Samuel Cahyawijaya. 2024. [Seacrowd: A multilingual multimodal data hub and benchmark suite for southeast asian languages](#).
- Sewon Min, Xinxu Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2022. [Rethinking the role of demonstrations: What makes in-context learning work?](#) In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 11048–11064,

- Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Peter Prettenhofer and Benno Stein. 2010. [Cross-language text classification using structural correspondence learning](#). In *Proceedings of the 48th Annual Meeting of the Association for Computational Linguistics*, pages 1118–1127, Uppsala, Sweden. Association for Computational Linguistics.
- Jack W. Rae, Sebastian Borgeaud, Trevor Cai, Katie Millican, Jordan Hoffmann, Francis Song, John Aslanides, Sarah Henderson, Roman Ring, Susannah Young, Eliza Rutherford, Tom Hennigan, Jacob Menick, Albin Cassirer, Richard Powell, George van den Driessche, Lisa Anne Hendricks, Mari-beth Rauh, Po-Sen Huang, Amelia Glaese, Johannes Welbl, Sumanth Dathathri, Saffron Huang, Jonathan Uesato, John Mellor, Irina Higgins, Antonia Creswell, Nat McAleese, Amy Wu, Erich Elsen, Siddhant Jayakumar, Elena Buchatskaya, David Budden, Esme Sutherland, Karen Simonyan, Michela Paganini, Laurent Sifre, Lena Martens, Xiang Lorraine Li, Adhiguna Kuncoro, Aida Nematzadeh, Elena Gribovskaya, Domenic Donato, Angeliki Lazaridou, Arthur Mensch, Jean-Baptiste Lespiau, Maria Tsim-poukelli, Nikolai Grigorev, Doug Fritz, Thibault Sotiaux, Mantas Pajarskas, Toby Pohlen, Zhitao Gong, Daniel Toyama, Cyprien de Masson d’Autume, Yujia Li, Tayfun Terzi, Vladimir Mikulik, Igor Babuschkin, Aidan Clark, Diego de Las Casas, Aurelia Guy, Chris Jones, James Bradbury, Matthew Johnson, Blake Hechtman, Laura Weidinger, Iason Gabriel, William Isaac, Ed Lockhart, Simon Osindero, Laura Rimell, Chris Dyer, Oriol Vinyals, Kareem Ayoub, Jeff Stanway, Lorraine Bennett, Demis Hassabis, Koray Kavukcuoglu, and Geoffrey Irving. 2022. [Scaling language models: Methods, analysis & insights from training gopher](#).
- Nils Reimers and Iryna Gurevych. 2019. [Sentence-bert: Sentence embeddings using siamese bert-networks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics.
- Nils Reimers and Iryna Gurevych. 2020. [Making monolingual sentence embeddings multilingual using knowledge distillation](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4512–4525, Online. Association for Computational Linguistics.
- David Rolnick, Arun Ahuja, Jonathan Schwarz, Timothy Lillicrap, and Gregory Wayne. 2019. [Experience replay for continual learning](#). In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.
- Teven Le Scao, Angela Fan, Christopher Akiki, El-lie Pavlick, Suzana Ilić, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François Yvon, Matthias Gallé, et al. 2022. Bloom: A 176b-parameter open-access multilingual language model. *arXiv preprint arXiv:2211.05100*.
- Holger Schwenk, Vishrav Chaudhary, Shuo Sun, Hongyu Gong, and Francisco Guzmán. 2021. [Wiki-Matrix: Mining 135M parallel sentences in 1620 language pairs from Wikipedia](#). In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 1351–1361, Online. Association for Computational Linguistics.
- Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. 2023. [Language models are multi-lingual chain-of-thought reasoners](#). In *The Eleventh International Conference on Learning Representations*.
- Shaden Smith, Mostofa Patwary, Brandon Norick, Patrick LeGresley, Samyam Rajbhandari, Jared Casper, Zhun Liu, Shrimai Prabhumoye, George Zerveas, Vijay Korthikanti, Elton Zhang, Rewon Child, Reza Yazdani Aminabadi, Julie Bernauer, Xia Song, Mohammad Shoeybi, Yuxiong He, Michael Houston, Saurabh Tiwary, and Bryan Catanzaro. 2022. [Using deepspeed and megatron to train megatron-turing nlG 530b, a large-scale generative language model](#).
- Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R. Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, Agnieszka Kluska, Aitor Lewkowycz, Akshat Agarwal, Alethea Power, Alex Ray, Alex Warstadt, Alexander W. Kocurek, Ali Safaya, Ali Tazarv, Alice Xiang, Alicia Parrish, Allen Nie, Aman Hussain, Amanda Askell, Amanda Dsouza, Ambrose Slone, Ammeet Rahane, Anantharaman S. Iyer, Anders Johan Andreassen, Andrea Madotto, Andrea Santilli, Andreas Stuhlmüller, Andrew M. Dai, Andrew La, Andrew Lampinen, Andy Zou, Angela Jiang, Angelica Chen, Anh Vuong, Animesh Gupta, Anna Gottardi, Antonio Norelli, Anu Venkatesh, Arash Gholamidavoodi, Arfa Tabassum, Arul Menezes, Arun Kirubara-jan, Asher Mullokandov, Ashish Sabharwal, Austin Herrick, Avia Efrat, Aykut Erdem, Ayla Karakaş, B. Ryan Roberts, Bao Sheng Loe, Barret Zoph, Bartłomiej Bojanowski, Batuhan Özyurt, Behnam Hedayatnia, Behnam Neyshabur, Benjamin Indén, Benno Stein, Berk Ekmekci, Bill Yuchen Lin, Blake Howald, Bryan Orinon, Cameron Diao, Cameron Dour, Catherine Stinson, Cedrick Argueta, Cesar Ferri, Chandan Singh, Charles Rathkopf, Chenlin Meng, Chitta Baral, Chiyu Wu, Chris Callison-Burch, Christopher Waites, Christian Voigt, Christopher D Manning, Christopher Potts, Cindy Ramirez, Clara E. Rivera, Clemencia Siro, Colin Raffel, Courtney Ashcraft, Cristina Garbacea, Damien Sileo, Dan Garrette, Dan Hendrycks, Dan Kilman, Dan Roth, C. Daniel Freeman, Daniel Khashabi, Daniel Levy, Daniel Moseguí González, Danielle Perszyk, Danny Hernandez, Danqi Chen, Daphne Ippolito, Dar Gilboa, David Dohan, David Drakard, David Jur-gens, Debajyoti Datta, Deep Ganguli, Denis Emelin,

Denis Kleyko, Deniz Yuret, Derek Chen, Derek Tam, Dieuwke Hupkes, Diganta Misra, Dilyar Buzan, Dimitri Coelho Mollo, Diyi Yang, Dong-Ho Lee, Dylan Schrader, Ekaterina Shutova, Ekin Dogus Cubuk, Elad Segal, Eleanor Hagerman, Elizabeth Barnes, Elizabeth Donoway, Ellie Pavlick, Emanuele Rodolà, Emma Lam, Eric Chu, Eric Tang, Erkut Erdem, Ernie Chang, Ethan A Chi, Ethan Dyer, Ethan Jerzak, Ethan Kim, Eunice Engefu Manyasi, Evgenii Zheltonozhskii, Fanyue Xia, Fatemeh Siar, Fernando Martínez-Plumed, Francesca Happé, François Chollet, Frieda Rong, Gaurav Mishra, Genta Indra Winata, Gerard de Melo, Germán Kruszewski, Giambattista Parascandolo, Giorgio Mariani, Gloria Xinyue Wang, Gonzalo Jaimovitch-Lopez, Gregor Betz, Guy Gur-Ari, Hana Galijasevic, Hannah Kim, Hannah Rashkin, Hannaneh Hajishirzi, Harsh Mehta, Hayden Bogar, Henry Francis Anthony Shevlin, Hinrich Schuetze, Hiromu Yakura, Hongming Zhang, Hugh Mee Wong, Ian Ng, Isaac Noble, Jaap Jumelet, Jack Geissinger, Jackson Kernion, Jacob Hilton, Jaehoon Lee, Jaime Fernández Fisac, James B Simon, James Koppel, James Zheng, James Zou, Jan Kocon, Jana Thompson, Janelle Wingfield, Jared Kaplan, Jarema Radom, Jascha Sohl-Dickstein, Jason Phang, Jason Wei, Jason Yosinski, Jekaterina Novikova, Jelle Bosscher, Jennifer Marsh, Jeremy Kim, Jeroen Taal, Jesse Engel, Jesujoba Alabi, Jiacheng Xu, Jiaming Song, Jillian Tang, Joan Waweru, John Burden, John Miller, John U. Balis, Jonathan Batchelder, Jonathan Berant, Jörg Froberg, Jos Rozen, Jose Hernandez-Orallo, Joseph Boudeman, Joseph Guerr, Joseph Jones, Joshua B. Tenenbaum, Joshua S. Rule, Joyce Chua, Kamil Kanclerz, Karen Livescu, Karl Krauth, Karthik Gopalakrishnan, Katerina Ignatyeva, Katja Markert, Kaustubh Dhole, Kevin Gimpel, Kevin Omondi, Kory Wallace Mathewson, Kristen Chiafullo, Ksenia Shkaruta, Kumar Shridhar, Kyle McDonell, Kyle Richardson, Laria Reynolds, Leo Gao, Li Zhang, Liam Dugan, Lianhui Qin, Lidia Contreras-Ochando, Louis-Philippe Morency, Luca Moschella, Lucas Lam, Lucy Noble, Ludwig Schmidt, Luheng He, Luis Oliveros-Colón, Luke Metz, Lütfi Kerem Senel, Maarten Bosma, Maarten Sap, Maartje Ter Hoeve, Maheen Farooqi, Manaal Faruqui, Mantas Mazeika, Marco Baturan, Marco Marelli, Marco Maru, Maria Jose Ramirez-Quintana, Marie Tolkiehn, Mario Giulianelli, Martha Lewis, Martin Potthast, Matthew L Leavitt, Matthias Hagen, Mátyás Schubert, Medina Orduna Baitemirova, Melody Arnaud, Melvin McElrath, Michael Andrew Yee, Michael Cohen, Michael Gu, Michael Ivanitskiy, Michael Starritt, Michael Strube, Michał Śwędrowski, Michele Bevilacqua, Michihiro Yasunaga, Mihir Kale, Mike Cain, Mimeo Xu, Mirac Suzgun, Mitch Walker, Mo Tiwari, Mohit Bansal, Moin Aminnaseri, Mor Geva, Mozhdah Gheini, Mukund Varma T, Nanyun Peng, Nathan Andrew Chi, Nayeon Lee, Neta Gur-Ari Krakover, Nicholas Cameron, Nicholas Roberts, Nick Doiron, Nicole Martinez, Nikita Nangia, Niklas Deckers, Niklas Muennighoff, Nitish Shirish Keskar, Niveditha S. Iyer, Noah Constant, Noah Fiedel, Nuan Wen, Oliver Zhang, Omar Agha, Omar El-

baghdadi, Omer Levy, Owain Evans, Pablo Antonio Moreno Casares, Parth Doshi, Pascale Fung, Paul Pu Liang, Paul Vicol, Pegah Alipoormolabashi, Peiyuan Liao, Percy Liang, Peter W Chang, Peter Eckersley, Phu Mon Htut, Pinyu Hwang, Piotr Miłkowski, Piyush Patil, Pouya Pezeshkpour, Priti Oli, Qiaozhu Mei, Qing Lyu, Qinlang Chen, Rabin Banjade, Rachel Etta Rudolph, Raefer Gabriel, Rahel Habacker, Ramon Risco, Raphaël Millièvre, Rhythm Garg, Richard Barnes, Rif A. Saurous, Riku Arakawa, Robbe Raymaekers, Robert Frank, Rohan Sikand, Roman Novak, Roman Sitelew, Roman Le Bras, Rosanne Liu, Rowan Jacobs, Rui Zhang, Russ Salakhutdinov, Ryan Andrew Chi, Seungjae Ryan Lee, Ryan Stovall, Ryan Teehan, Rylan Yang, Sahib Singh, Saif M. Mohammad, Sajant Anand, Sam Dillavou, Sam Shleifer, Sam Wiseman, Samuel Gruetter, Samuel R. Bowman, Samuel Stern Schoenholz, Sanghyun Han, Sanjeev Kwatra, Sarah A. Rous, Sarik Ghazarian, Sayan Ghosh, Sean Casey, Sebastian Bischoff, Sebastian Gehrmann, Sebastian Schuster, Sepideh Sadeghi, Shadi Hamdan, Sharon Zhou, Shashank Srivastava, Sherry Shi, Shikhar Singh, Shima Asaadi, Shixiang Shane Gu, Shubh Pachchigar, Shubham Toshniwal, Shyam Upadhyay, Shyamolima Shammie Debnath, Siamak Shakeri, Simon Thormeyer, Simone Melzi, Siva Reddy, Sneha Priscilla Makini, Soo-Hwan Lee, Spencer Torene, Sriharsha Hatwar, Stanislas Dehaene, Stefan Divic, Stefano Ermon, Stella Biderman, Stephanie Lin, Stephen Prasad, Steven Piantadosi, Stuart Shieber, Summer Mishnerghi, Svetlana Kiritchenko, Swaroop Mishra, Tal Linzen, Tal Schuster, Tao Li, Tao Yu, Tariq Ali, Tatsunori Hashimoto, Te-Lin Wu, Théo Desbordes, Theodore Rothschild, Thomas Phan, Tianle Wang, Tiberius Nkinyili, Timo Schick, Timofei Kornev, Titus Tunduny, Tobias Gerstenberg, Trenton Chang, Trishala Neeraj, Tushar Khot, Tyler Shultz, Uri Shaham, Vedant Misra, Vera Demberg, Victoria Nyamai, Vikas Raunak, Vinay Venkatesh Ramasesh, vinay uday prabhu, Vishakh Padmakumar, Vivek Srikumar, William Fedus, William Saunders, William Zhang, Wout Vossen, Xiang Ren, Xiaoyu Tong, Xinran Zhao, Xinyi Wu, Xudong Shen, Yadollah Yaghoobzadeh, Yair Lakretz, Yangqiu Song, Yasaman Bahri, Yejin Choi, Yichi Yang, Yiding Hao, Yifu Chen, Yonatan Belinkov, Yu Hou, Yufang Hou, Yuntao Bai, Zachary Seid, Zhuoye Zhao, Zijian Wang, Zijie J. Wang, Zirui Wang, and Ziyi Wu. 2023. [Beyond the imitation game: Quantifying and extrapolating the capabilities of language models.](#) *Transactions on Machine Learning Research*.

Eshaan Tanwar, Subhabrata Dutta, Manish Borthakur, and Tanmoy Chakraborty. 2023. [Multilingual LLMs are better cross-lingual in-context learners with alignment.](#) In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6292–6307, Toronto, Canada. Association for Computational Linguistics.

MosaicML NLP Team. 2023. [Introducing mpt-7b: A](#)

[new standard for open-source, commercially usable llms](#). Accessed: 2023-05-05.

- NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Hefernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, Anna Sun, Skyler Wang, Guillaume Wenzek, Al Youngblood, Bapi Akula, Loic Barrault, Gabriel Mejia Gonzalez, Prangthip Hansanti, John Hoffman, Semarley Jarrett, Kaushik Ram Sadagopan, Dirk Rowe, Shannon Spruit, Chau Tran, Pierre Andrews, Necip Fazil Ayan, Shruti Bhosale, Sergey Edunov, Angela Fan, Cynthia Gao, Vedanuj Goswami, Francisco Guzmán, Philipp Koehn, Alexandre Mourachko, Christophe Ropers, Safiyyah Saleem, Holger Schwenk, and Jeff Wang. 2022. No language left behind: Scaling human-centered machine translation.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, brian ichter, Fei Xia, Ed H. Chi, Quoc V Le, and Denny Zhou. 2022. [Chain of thought prompting elicits reasoning in large language models](#). In *Advances in Neural Information Processing Systems*.
- Haryo Akbarianto Wibowo, Erland Hilman Fuadi, Made Nindyatama Nityasya, Radityo Eko Prasajo, and Alham Fikri Aji. 2023. [Copal-id: Indonesian language reasoning with local culture and nuances](#).
- Bryan Wilie, Karissa Vincentio, Genta Indra Winata, Samuel Cahyawijaya, Xiaohong Li, Zhi Yuan Lim, Sidik Soleman, Rahmad Mahendra, Pascale Fung, Syafri Bahar, and Ayu Purwarianti. 2020. [IndoNLU: Benchmark and resources for evaluating Indonesian natural language understanding](#). In *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing*, pages 843–857, Suzhou, China. Association for Computational Linguistics.
- Genta Winata, Shijie Wu, Mayank Kulkarni, Thamar Solorio, and Daniel Preotiuc-Pietro. 2022a. [Cross-lingual few-shot learning on unseen languages](#). In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 777–791, Online only. Association for Computational Linguistics.
- Genta Indra Winata, Alham Fikri Aji, Samuel Cahyawijaya, Rahmad Mahendra, Fajri Koto, Ade Romadhony, Kemal Kurniawan, David Moeljadi, Radityo Eko Prasajo, Pascale Fung, et al. 2022b. Nusax: Multilingual parallel sentiment dataset for 10 indonesian local languages. *arXiv preprint arXiv:2205.15960*.
- Genta Indra Winata, Samuel Cahyawijaya, Zihan Liu, Zhaojiang Lin, Andrea Madotto, and Pascale Fung. 2021a. [Are multilingual models effective in code-switching?](#) In *Proceedings of the Fifth Workshop on Computational Approaches to Linguistic Code-Switching*, pages 142–153, Online. Association for Computational Linguistics.
- Genta Indra Winata, Andrea Madotto, Zhaojiang Lin, Rosanne Liu, Jason Yosinski, and Pascale Fung. 2021b. Language models are few-shot multilingual learners. In *Proceedings of the 1st Workshop on Multilingual Representation Learning*, pages 1–15.
- Genta Indra Winata, Andrea Madotto, Zhaojiang Lin, Rosanne Liu, Jason Yosinski, and Pascale Fung. 2021c. [Language models are few-shot multilingual learners](#). In *Proceedings of the 1st Workshop on Multilingual Representation Learning*, pages 1–15, Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. 2022. [An explanation of in-context learning as implicit bayesian inference](#). In *International Conference on Learning Representations*.
- Zheng-Xin Yong, Hailey Schoelkopf, Niklas Muenighoff, Alham Fikri Aji, David Ifeoluwa Adelani, Khalid Almubarak, M Saiful Bari, Lintang Sutawika, Jungo Kasai, Ahmed Baruwu, Genta Indra Winata, Stella Biderman, Dragomir Radev, and Vassilina Nikoulina. 2022. [Bloom+1: Adding language support to bloom for zero-shot prompting](#).
- Zheng-Xin Yong, Ruochen Zhang, Jessica Zosa Forde, Skyler Wang, Samuel Cahyawijaya, Holy Lovenia, Genta Indra Winata, Lintang Sutawika, Jan Christian Blaise Cruz, Long Phan, et al. 2023. Prompting multilingual large language models to generate code-mixed texts: The case of south east asian languages. *arXiv preprint arXiv:2303.13592*.
- Ruochen Zhang, Samuel Cahyawijaya, Jan Christian Blaise Cruz, and Alham Fikri Aji. 2023. [Multilingual large language models are not \(yet\) code-switchers](#).

A Languages Under Study

We conduct experiments on 32 languages, with 25 low-resource languages and 7 relatively high-resource languages. We provide the detailed list of all the languages under study in Table 3.

Language Code	Language Name	Dataset Name	Test Size	Geographic Region	Language Family	%BLOOM Pretraining	%XGLM Pretraining
btk	Batak	NusaTranslation	1200	South-East Asia	Austronesian	-	-
sun	Sundanese	NusaTranslation	1200	South-East Asia	Austronesian	-	-
jav	Javanese	NusaTranslation	1200	South-East Asia	Austronesian	-	-
mad	Madurese	NusaTranslation	1200	South-East Asia	Austronesian	-	-
mak	Makassarese	NusaTranslation	1200	South-East Asia	Austronesian	-	-
min	Minangkabau	NusaTranslation	1200	South-East Asia	Austronesian	-	-
amh	Amharic	MasakhaNews	376	Africa	Afro-Asiatic	-	-
hau	Hausa	MasakhaNews	637	Africa	Afro-Asiatic	-	-
ibo	Igbo	MasakhaNews	390	Africa	Niger-Congo	0.00%	-
lug	Luganda	MasakhaNews	223	Africa	Niger-Congo	0.00%	-
pcm	Nigerian Pidgin	MasakhaNews	305	Africa	English Creole	-	-
sna	chiShona	MasakhaNews	369	Africa	Niger-Congo	-	-
swa	Kiswahili	MasakhaNews	476	Africa	Niger-Congo	0.01%	0.25%
xho	isiXhosa	MasakhaNews	297	Africa	Niger-Congo	0.00%	-
yor	Yorùbá	MasakhaNews	411	Africa	Niger-Congo	0.01%	-
aym	Aymara	AmericasNLI	750	South America	Aymaran	-	-
bzd	Bribri	AmericasNLI	750	South America	Chibchan	-	-
cni	Asháninka	AmericasNLI	750	South America	Arawak	-	-
grn	Guaraní	AmericasNLI	750	South America	Tupian	-	-
hch	Wixarika	AmericasNLI	750	South America	Uto-Aztecan	-	-
nah	Nahuatl	AmericasNLI	738	South America	Uto-Aztecan	-	-
oto	Otomí	AmericasNLI	748	South America	Oto-Manguean	-	-
quy	Quechua	AmericasNLI	750	South America	Quechuan	-	0.01%
shp	Shipibo-Konibo	AmericasNLI	750	South America	Pano-Tacanan	-	-
tar	Rarámuri	AmericasNLI	750	South America	Uto-Aztecan	-	-
arb	Arabic	TweetSentimentMultilingual	870	Northern Africa	Afro-Asiatic	4.64%	0.75%
fra	French	TweetSentimentMultilingual	870	Europe	Indo-European	12.90%	3.00%
deu	German	TweetSentimentMultilingual	870	Europe	Indo-European	-	3.50%
hin	Hindi	TweetSentimentMultilingual	870	Central Asia	Indo-European	1.53%	1.00%
ita	Italian	TweetSentimentMultilingual	870	Europe	Indo-European	-	1.50%
por	Portuguese	TweetSentimentMultilingual	870	Europe	Indo-European	4.91%	2.25%
spa	Spanish	TweetSentimentMultilingual	870	Europe	Indo-European	10.85%	3.25%

Table 3: List of languages under study. "-" denotes the language is not on the pre-training dataset, while 0.00% denotes a very small percentage (<0.01%) of the pre-training data is in that language.

B Alignment Prompt

We showcase the example prompt for cross-lingual in-context learning, in-context label alignment, and in-context query alignment in Figure 10.

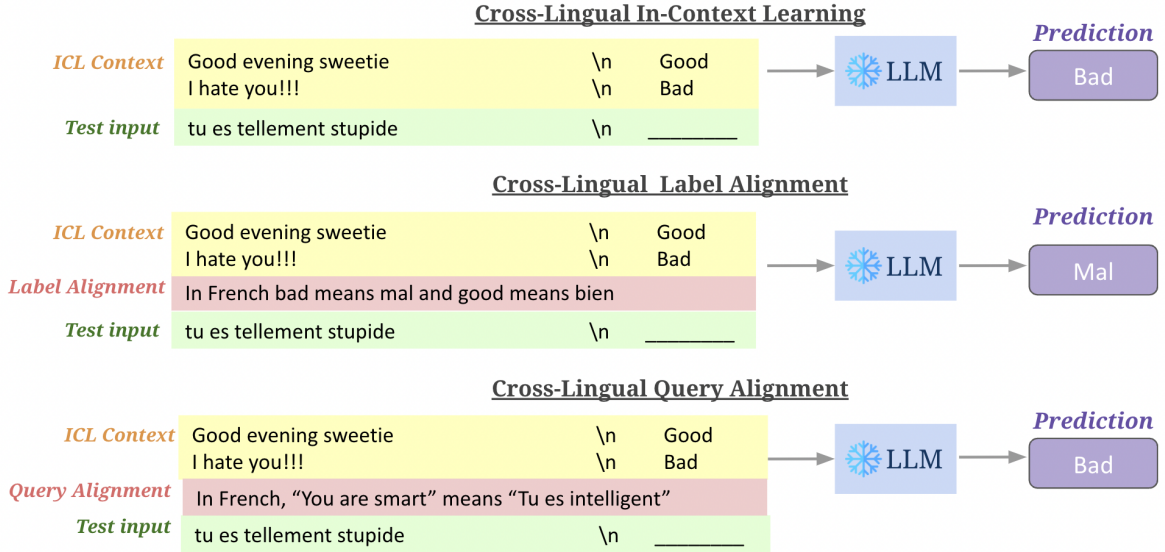


Figure 10: Example prompt for in-context label alignment and in-context query alignment.

C Analysis on Cross-lingual Semantic Similarity

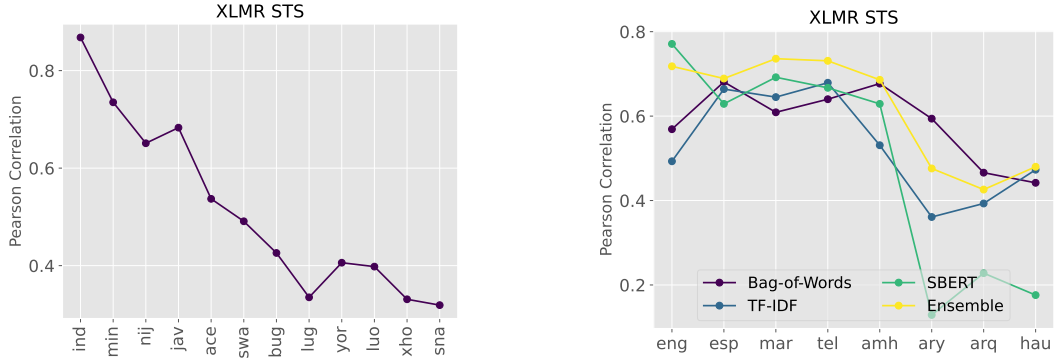


Figure 11: **(top)** Correlation of cross-lingual similarity with the correct label for the XLMR STS model. **(bottom)** Correlation of monolingual similarity with the correct label for the XLMR STS model.

We showcase that the semantic representation for these languages might not be well aligned with the high-resource languages. We construct sentence similarity dataset covering 26 languages by utilizing the translation samples from two machine translation datasets, i.e., MAFAND (Adelani et al., 2022a) and NusaX-MT (Winata et al., 2022b). We create a balanced dataset with 50% positive pairs and 50% negative pairs over all 26 languages. We measure the cross-lingual semantic similarity performance of using Sentence Transformers (Reimers and Gurevych, 2019). We also conduct monolingual semantic similarity analysis for various languages using the data from SemEval 2024 Task 12: Textual Semantic Relatedness dataset⁵. For the monolingual semantic similarity we add additional word frequency features including bag-of-words and TF-IDF to improve the retrieval quality of the semantic similarity model.

As shown in Figure 11, both monolingual and cross-lingual semantic similarity on more low-resource languages are generally yield a much lower correlation which signifies the limitation of the sentence

⁵https://github.com/semantic-textual-relatedness/Semantic_Relatedness_SemEval2024

embedding model to represent the sentences on these languages. Nevertheless, for monolingual semantic similarity, it is possible to improve the similarity on these low-resource languages with minimal trade off on the other language by employing character/word frequency features to support the semantic similarity model. With that in mind, we explore an alternative approach for cross-lingual retrieval by using monolingual semantic similarity and an external parallel corpus. We called this semantic similarity method as **translation semantic similarity**. The comparison of cross-lingual retrieval using cross-lingual semantic similarity and translation semantic similarity is shown in Figure 12.

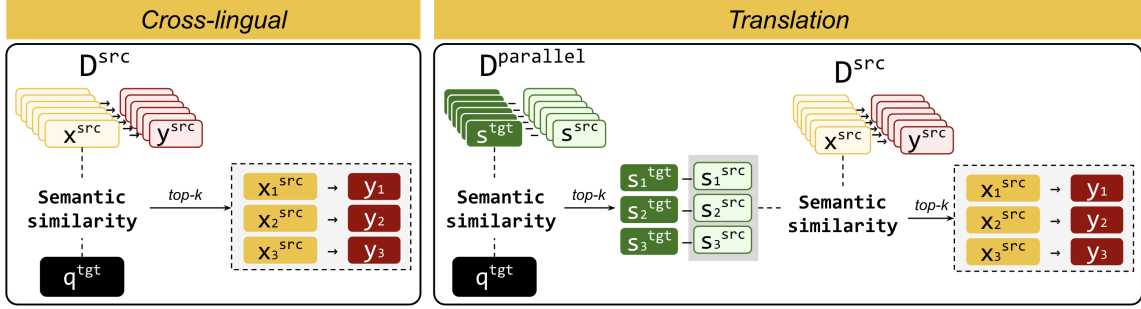


Figure 12: We explore two semantic similarity methods for cross-lingual exemplar retrieval in X-ICL, i.e., cross-lingual semantic similarity and translation semantic similarity (T-ICL).

Language	Label Set						
eng	business	entertainment	health	politics	religion	sports	technology
hau	kasuwanci	nishadi	lafiya	siyasa	addini	wasanni	fasaha
ibo	azumahia	nturundu	ahuike	ndoro ndoro ochichi	okpukpere chi	egwuregwu	teknuzu
lug	bizinensi	okwesanyusa	obulamu	ebyobufuzi	eddiini	ebyemizannyo	tekinolojiya
pcm	business	entertainment	health	politics	religion	sports	technology
sna	business	varaidzo	utano	zvematongerwo enyika	chitendero	mitambo	teknolojia
swa	biashara	burudani	afya	siasa	dini	michezo	teknolojia
xho	ishishini	ukuzonwabisa	impilo	kwezopolitiko	unqulo	ezemidlalo	iteknoloji
yor	işowo	Idanilaraya	ilera	oselu	esin	idaraya	ona ero

Table 4: Label set for each language of the MasakhaNews dataset.

D Language Label

We provide the label set in the source and target languages used in all the languages under study in MasakhaNews, NusaTranslation, AmericasNLI, and TweetSentimentMultilingual on Table 4, Table 6, Table 7, and Table 5, respectively.

Language	Label Set		
eng	negative	neutral	positive
fra	négatif	neutre	positif
deu	negativ	neutral	positiv
ita	negativo	neutro	positivo
por	negativo	neutro	positivo
spa	negativo	neutral	positivo

Table 5: Label set for each language in the TweetSentimentMultilingual dataset.

Language	Label Set		
eng	negative	neutral	positive
ind	negatif	netral	positif
btk	negatif	netral	positif
sun	negatif	netral	positif
jav	negatif	netral	positif
mad	negatif	netral	positif
mak	negatif	netral	positif
min	negatif	netral	positif

Table 6: Label set for each language of the NusaTranslation dataset.

Language	Label Set		
eng	entailment	neutral	contradiction
spa	vinculación	neutral	contradicción
aym	vinculación	niwtrala	contradicción
bzd	-	-	-
cni	-	-	-
grn	vinculación	ñemombyte	contradicción
hch	-	-	-
nah	-	-	-
oto	vinculación	neutral	contradicción
quy	hukllanakuy	chawpi	contradicción
shp	-	-	-
tar	-	-	-

Table 7: Label set for each language of the AmericasNLI dataset.

E Effect of Machine Translation Quality to X-ICL

Dataset	Language	Language	chrF++ (xxx2eng)	XGLM		BLOOMZ	
	Code	Name		Zero-Shot (MT)	ICL (MT)	Zero-Shot (MT)	ICL (MT)
NusaTranslation	min	Minangkabau	60.30	68.32	67.28	67.26	76.83
NusaTranslation	sun	Sundanese	60.7	71.58	70.78	76.31	80.53
NusaTranslation	jav	Javanese	61.4	71.26	68.35	73.89	78.95
AmericasNLI	aym	Aymara	31.7	16.94	34.52	16.66	35.8
AmericasNLI	quy	Quechua	32.7	16.66	37.24	16.66	39.19
AmericasNLI	grn	Guaraní	47.6	16.66	34.42	16.66	37.79
TweetSentiMulti	spa	Spanish	58.3	42.14	45.38	45.47	55.8
TweetSentiMulti	ita	Italian	60.6	39.61	43.39	45.04	54.51
TweetSentiMulti	arb	Arabic	64.6	33.97	50.66	35.73	55.28
TweetSentiMulti	hin	Hindi	65.	32.11	40.43	35.09	45.40
TweetSentiMulti	deu	German	66.70	36.37	45.07	42.98	51.10
TweetSentiMulti	fra	French	67.20	36.91	41.87	40.22	55.73
TweetSentiMulti	por	Portuguese	70.60	39.04	45.02	42.21	53.42
MasakhaNews	yor	Yorùbá	43.80	45.69	74.62	75.42	81.64
MasakhaNews	lug	Luganda	44.90	34.71	59.98	70.54	62.82
MasakhaNews	sna	chiShona	49.20	60.53	72.80	68.71	73.85
MasakhaNews	ibo	Igbo	52.50	44.32	73.79	71.69	77.21
MasakhaNews	hau	Hausa	55.30	43.99	59.74	67.30	67.19
MasakhaNews	amh	Amharic	58.10	62.88	81.40	82.73	84.92
MasakhaNews	xho	isiXhosa	58.50	33.41	65.66	58.36	63.30
MasakhaNews	swa	Kiswahili	63.50	52.03	67.10	75.49	71.42
Pearson Correlation w/ chrF++				0.416	0.102	0.247	0.238

Table 8: Performance of NLLB 1.3B model on FLORES-200 with the machine-translated zero-shot and few-shot ICL performance of XGLM and BLOOMZ using the corresponding NLLB translation.

We showcase that the MT model performance plays a huge role in determining the language understanding quality through machine translation (MT). We showcase the MT model performance on the devtest subset of FLORES-200 (Goyal et al., 2022) along with the zero-shot with MT and few-shot ICL with MT performance in Table 8. The zero-shot (MT) performance has a low-to-moderate correlation with the machine translation quality (chrF++) of the model (0.416 for XGLM and 0.247 for BLOOMZ), while the few-shot ICL (MT) has a lower correlation (0.102 for XGLM and 0.238 for BLOOMZ) potentially due to the effect of other factors such as the semantic similarity exemplar selection and the quality of the ICL data itself. Our result indicates that, despite being effective for language understanding, the MT-based zero-shot and few-shot inference approach depends on the quality of the machine translation models. Moreover, an MT-based solution might not work as well for cultural-specific tasks which have been addressed in various works (Kabra et al., 2023; Koto et al., 2023; Wibowo et al., 2023).

F Effects of Source Languages

We explore alternative source languages for NusaTranslation (Cahyawijaya et al., 2023b). For NusaTranslation we utilize Indonesian as the source language because Indonesian is the closely related to the languages under study on the corresponding dataset and is widely spoken languages in the respective region. We modify both the prompt language and the source ICL dataset D^{src} .

The result is shown in Figure 13. We can clearly see that in most cases, using English as the source language tends to produce better score than these closely related languages. Similar observation is also reported in prior works (Cahyawijaya et al., 2023a; Asai et al., 2023) which evaluates the prompt using different prompt language. Our experiment further extend the generalization to the X-ICL setting, where X-ICL using English exemplars outperforms X-ICL with a more closely related languages exemplars.

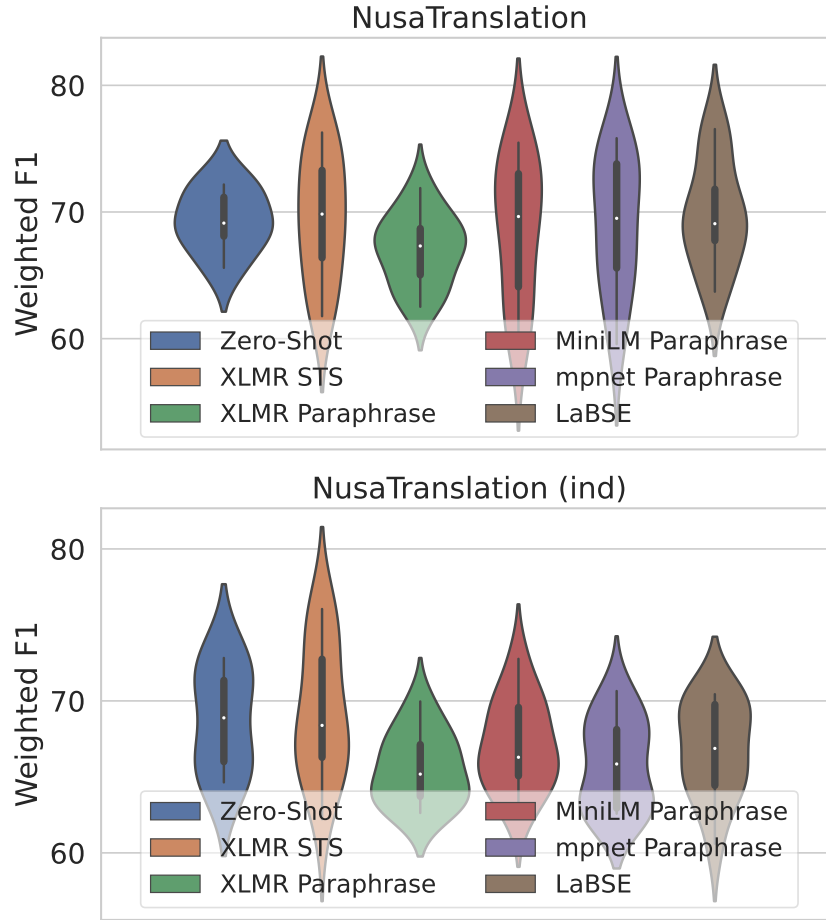


Figure 13: Performance of BLOOM-7B1 on NusaTranslation using **(top)** English prompt with English ICL exemplars and **(bottom)** Indonesian prompt with Indonesian ICL exemplars.

G Visualization of BLOOM Result

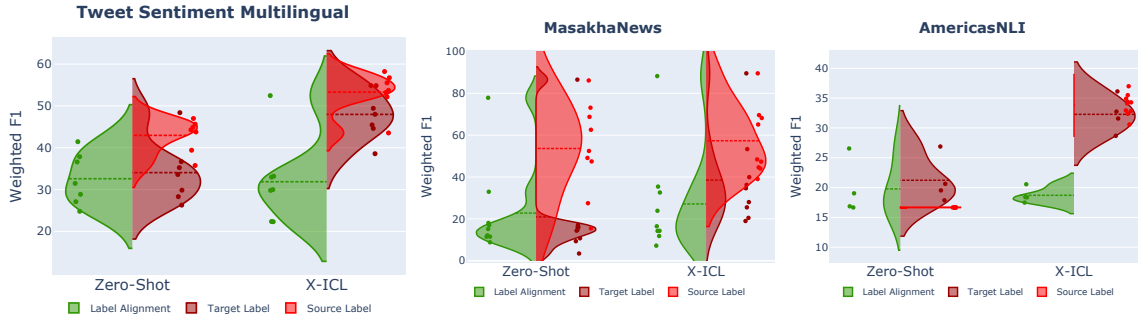


Figure 14: Performance of BLOOM-7B1 with in-context label alignment, target-only label, and source-only label on (left) higher-resource, (center) low-resource African, and (right) low-resource American languages.

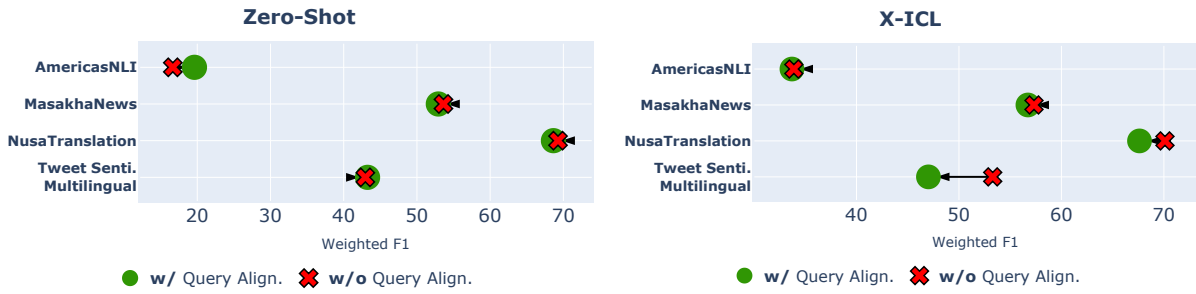


Figure 15: Performance of BLOOM-7B1 with and without query alignment on (left) higher-resource, (center) low-resource African, and (right) low-resource American languages.

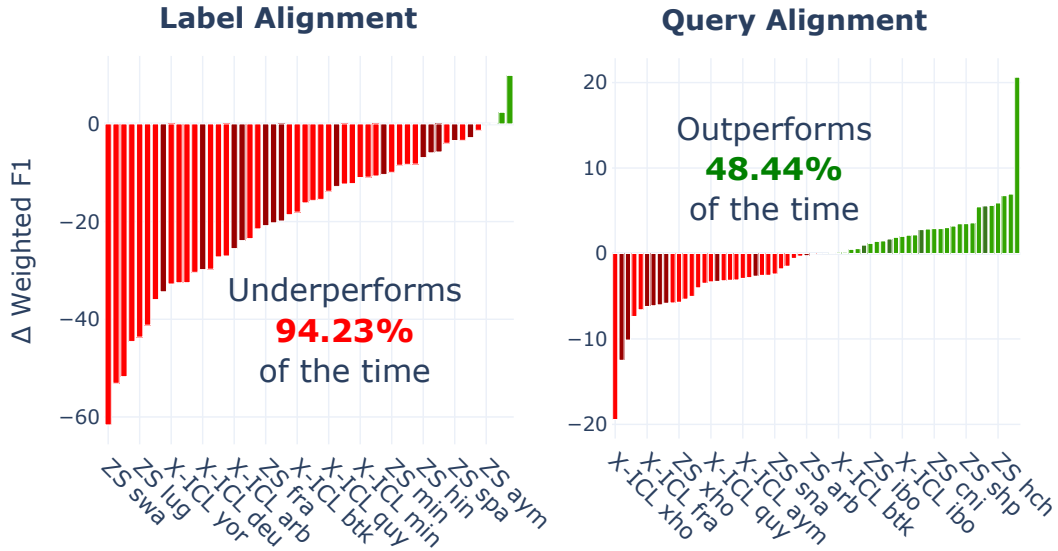


Figure 16: Δ Weighted F1 of (left) in-context label alignment and (right) in-context query alignment against non-alignment baseline. A score < 0 indicates the in-context alignment degrades the performance.

H Detailed Per Dataset Results

The detailed the main results for each different inference type for XGLM-7.5B in Table 9, Table 10, Table 11, and Table 12 for TweetSentimentMultilingual MasakhaNews, NusaTranslation, AmericasNLI,

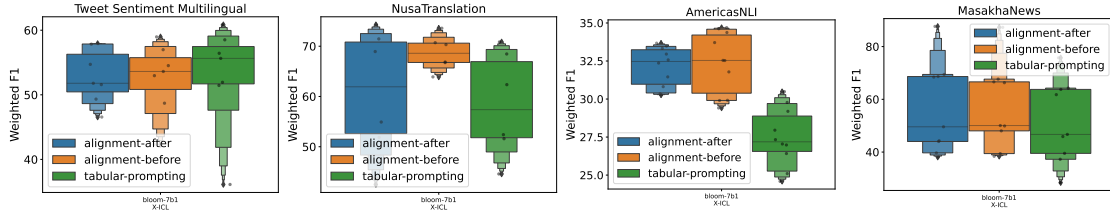


Figure 17: Performance of BLOOM-7B1 with different alignment formats ordered by the degree of formatting consistency on (1) higher-resource languages, (2) low-resource Indonesian languages, (3) low-resource American languages, and (4) low-resource African languages.

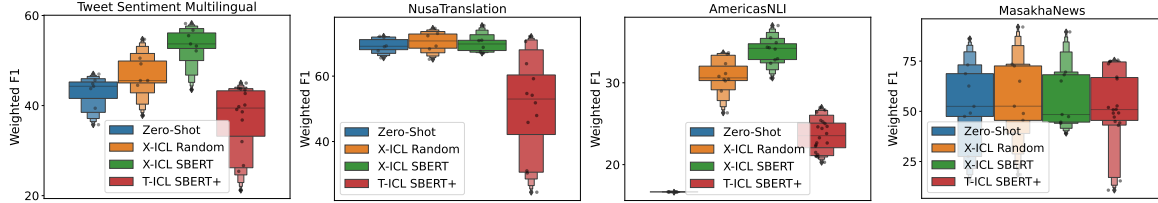


Figure 18: Performance of BLOOM-7B1 with different in-context learning retrievals covering monolingual, cross-lingual, translation semantic similarity on (1) higher-resource languages, (2) low-resource Indonesian languages, (3) low-resource American languages, and (4) low-resource African languages.

respectively. The detailed results for each different inference type for BLOOM-7B1 in Table 13, Table 14, Table 15, and Table 16 for TweetSentimentMultilingual MasakhaNews, NusaTranslation, AmericasNLI, respectively.

Inference Type	arb	deu	fra	hin	ita	por	spa
Zero-Shot							
Source-Only Label	38.19	39.82	32.82	32.09	38.39	44.83	51.04
+ Query Alignment	38.28	43.28	40.27	34.18	38.58	43.39	42.15
Target-Only Label	34.44	48.99	39.86	19.12	42.25	36.85	47.88
Label Alignment	21.15	35.41	27.97	28.50	31.85	28.82	30.18
Zero-Shot (MT)	33.97	36.37	36.91	32.11	39.61	39.04	42.14
ICL Random	40.39	38.94	36.50	36.16	37.10	36.85	44.66
ICL SBERT	46.60	45.56	54.04	34.02	48.43	51.01	45.90
ICL SBERT (MT)	50.66	45.07	41.87	40.43	43.39	45.02	45.38
X-ICL Random	35.53	40.16	37.38	32.49	40.98	39.46	39.83
X-ICL SBERT							
Source-Only Label	49.21	47.35	42.80	38.15	47.24	48.01	47.67
+ Query Alignment	45.28	48.85	46.35	39.62	44.83	50.65	44.20
Target-Only Label	47.88	45.05	43.37	37.23	42.99	46.40	42.58
Label Alignment	29.51	27.15	37.97	30.10	44.50	40.91	31.96

Table 9: Experiment results for XGLM-7.5B on TweetSentimentMultilingual dataset. "-" denotes the experiment is not conducted due to no machine translation system is available.

Inference Type	amh	hau	ibo	lug	pcm	sna	swa	xho	yor
Zero-Shot									
Source-Only Label	17.62	32.64	51.11	22.80	57.07	42.66	49.93	28.60	54.96
+ Query Alignment	25.79	36.81	59.07	39.51	72.65	42.68	58.12	21.97	48.28
Target-Only Label	11.92	7.36	3.72	13.94	58.59	15.25	53.29	2.11	12.88
Label Alignment	10.19	7.08	4.19	14.40	60.63	19.18	47.46	22.12	17.80
Zero-Shot (MT)	62.88	43.99	44.32	34.71	56.73	60.53	52.03	33.41	45.69
ICL Random	20.36	37.92	63.33	38.93	83.01	43.03	65.62	49.26	65.65
ICL SBERT	60.75	61.39	69.86	48.23	93.02	59.56	73.07	43.79	70.84
ICL SBERT (MT)	81.40	59.74	73.79	59.98	87.20	72.80	67.10	65.66	74.62
X-ICL Random	24.11	38.01	62.32	46.23	85.38	51.70	58.98	47.79	66.77
X-ICL SBERT									
Source-Only Label	55.18	37.08	64.28	46.95	88.27	41.87	63.05	49.10	65.74
+ Query Alignment	51.43	40.53	62.05	44.70	86.61	44.58	65.59	40.27	57.24
Target-Only Label	53.06	19.19	27.87	27.99	88.59	25.49	37.02	25.71	31.64
Label Alignment	10.19	13.79	6.40	12.35	90.88	12.71	62.73	21.41	16.00

Table 10: Experiment results for XGLM-7.5B on MasakhaNews dataset. "-" denotes the experiment is not conducted due to no machine translation system is available. SBERT denotes exemplar selection using a semantic similarity model.

Inference Type	btk	jav	mad	mak	min	sun
Zero-Shot						
Source-Only Label	58.60	62.92	60.75	55.90	64.29	63.67
+ Query Alignment	52.60	62.77	56.74	49.50	63.51	57.08
Target-Only Label	41.63	47.52	45.85	47.53	42.20	43.12
Label Alignment	30.58	28.17	31.81	37.79	28.59	29.83
Zero-Shot (MT)	-	71.26	-	60.68	68.32	71.58
ICL Random	59.68	60.85	59.28	60.83	62.91	59.52
ICL SBERT	59.59	60.84	60.81	62.39	66.11	61.68
ICL SBERT (MT)	-	68.35	-	59.61	67.28	70.78
X-ICL Random	60.54	61.74	63.02	58.21	63.87	61.66
X-ICL SBERT						
Source-Only Label	60.69	62.83	59.78	60.30	63.95	62.44
+ Query Alignment	52.41	60.38	53.06	52.77	61.36	56.62
Target-Only Label	56.02	59.57	56.83	48.61	64.80	59.35
Label Alignment	55.60	61.13	57.45	55.10	62.52	58.42

Table 11: Experiment results for XGLM-7.5B on NusaTranslation dataset. "-" denotes the experiment is not conducted due to no machine translation system is available. SBERT denotes exemplar selection using a semantic similarity model.

Inference Type	aym	bzd	cni	grn	hch	nah	oto	quy	shp	tar
Zero-Shot										
Source-Only Label	16.68	16.66	16.66	16.61	17.68	18.88	19.31	16.66	17.62	16.66
+ Query Alignment	29.56	30.79	28.04	29.15	32.07	33.05	32.23	33.77	32.33	30.82
Target-Only Label	19.88	-	-	17.79	-	-	17.69	22.63	-	-
Label Alignment	22.31	-	-	17.90	-	-	25.52	29.17	-	-
Zero-Shot (MT)	16.94	-	-	16.66	-	-	-	16.66	-	-
ICL Random	32.43	28.66	30.42	29.91	29.15	32.70	29.63	32.98	30.28	31.74
ICL SBERT	34.65	28.26	30.62	34.34	31.10	33.89	28.02	32.64	28.90	30.97
ICL SBERT (MT)	34.52	-	-	34.42	-	-	-	37.24	-	-
X-ICL Random	28.96	32.55	30.72	28.95	33.01	33.55	28.88	34.78	32.16	31.43
X-ICL SBERT										
Source-Only Label	33.20	33.99	31.99	33.88	31.00	30.80	30.97	34.24	26.95	32.74
+ Query Alignment	35.30	32.83	35.60	32.71	33.04	28.05	31.02	34.29	30.57	32.97
Target-Only Label	30.58	-	-	34.76	-	-	31.19	28.32	-	-
Label Alignment	25.30	-	-	17.37	-	-	25.79	25.61	-	-

Table 12: Experiment results for XGLM-7.5B on AmericasNLI dataset. "-" denotes the experiment is not conducted due to no machine translation system is available.

Inference Type	arb	deu	fra	hin	ita	por	spa
Zero-Shot							
Source-Only Label	43.77	39.40	45.62	35.75	46.98	44.28	44.84
+ Query Alignment	43.51	40.38	42.96	37.45	40.98	49.85	47.64
Target-Only Label	33.59	28.30	29.85	26.28	36.68	35.23	48.37
Label Alignment	37.86	36.60	24.80	28.86	27.10	31.47	41.44
Zero-Shot (MT)	35.73	42.98	40.22	35.09	45.04	42.21	45.47
ICL Random	41.71	50.44	37.72	37.24	49.86	48.58	51.10
ICL SBERT	51.17	55.83	57.67	38.27	51.81	57.68	60.28
ICL SBERT (MT)	55.28	51.10	55.73	45.40	54.51	53.42	55.83
X-ICL Random	44.50	45.54	45.56	37.79	50.52	49.26	54.71
X-ICL SBERT							
Source-Only Label	55.52	52.14	53.22	43.53	53.69	58.20	56.73
+ Query Alignment	45.38	46.03	47.01	43.65	41.19	52.38	53.46
Target-Only Label	44.61	47.99	45.45	38.57	54.85	54.84	49.41
Label Alignment	29.99	22.32	32.98	33.18	29.80	52.45	22.36

Table 13: Experiment results for BLOOM-7B1 model on TweetSentimentMultilingual dataset. "-" denotes the experiment is not conducted due to no machine translation system is available.

Inference Type	amh	hau	ibo	lug	pcm	sna	swa	xho	yor
Zero-Shot									
Source-Only Label	15.47	47.45	62.58	52.46	86.14	49.12	73.12	27.49	68.75
+ Query Alignment	43.33	45.99	71.56	51.43	89.22	38.83	71.87	24.61	73.66
Target-Only Label	10.72	9.34	17.11	14.36	86.53	16.18	14.73	3.44	15.97
Label Alignment	12.10	11.48	18.04	8.81	77.86	32.95	11.49	15.18	17.01
Zero-Shot (MT)	82.73	67.30	71.69	70.54	84.50	68.71	75.49	58.36	75.42
ICL Random	26.74	42.29	73.91	45.04	85.46	49.53	73.59	36.86	72.71
ICL SBERT	61.84	60.77	79.24	49.86	92.19	66.67	74.57	43.63	79.28
ICL SBERT (MT)	84.92	67.19	77.21	62.82	90.23	73.85	71.42	63.30	81.64
X-ICL Random	18.57	45.50	72.59	48.92	91.98	52.54	64.99	38.98	73.45
X-ICL SBERT									
Source-Only Label	47.35	39.04	69.56	48.41	89.54	44.62	65.10	44.04	68.20
+ Query Alignment	36.09	42.43	63.76	45.88	84.34	46.71	69.99	21.78	65.26
Target-Only Label	53.33	18.98	36.25	25.50	89.54	28.02	39.94	20.47	34.63
Label Alignment	23.87	11.82	16.45	7.20	88.23	14.23	32.60	14.29	35.42

Table 14: Experiment results for BLOOM-7B1 model on MasakhaNews dataset. "-" denotes the experiment is not conducted due to no machine translation system is available.

Inference Type	btk	jav	mad	mak	min	sun
Zero-Shot						
Source-Only Label	65.58	69.00	67.78	69.24	72.17	71.80
+ Query Alignment	65.50	71.13	67.20	61.87	72.14	73.98
Target-Only Label	66.76	68.22	67.22	64.21	67.79	68.12
Label Alignment	61.62	60.77	59.31	58.57	62.25	60.89
Zero-Shot (MT)	-	73.89	-	57.87	67.26	76.31
ICL Random	65.68	68.40	65.33	62.84	70.45	65.32
ICL SBERT	62.84	72.70	64.38	61.77	76.27	75.04
ICL SBERT (MT)	-	78.95	-	67.56	76.83	80.53
X-ICL Random	68.32	73.05	69.17	65.15	74.60	72.33
X-ICL SBERT						
Source-Only Label	67.04	70.86	68.79	67.49	75.45	70.97
+ Query Alignment	67.18	68.30	66.24	64.39	70.10	69.47
Target-Only Label	59.99	69.00	62.48	61.28	72.53	71.40
Label Alignment	48.97	43.81	47.30	34.98	64.47	55.31

Table 15: Experiment results for BLOOM-7B1 model on NusaTranslation dataset. "-" denotes the experiment is not conducted due to no machine translation system is available.

Inference Type	aym	bzd	cni	grn	hch	nah	oto	quy	shp	tar
Zero-Shot										
Source-Only Label	16.66	16.66	16.66	16.66	16.66	16.66	16.62	16.66	16.66	16.66
+ Query Alignment	19.87	18.07	19.57	18.13	22.57	19.58	18.51	19.52	20.15	20.14
Target-Only Label	26.88	-	-	19.52	-	-	17.86	20.62	-	-
Label Alignment	16.66	-	-	19.03	-	-	26.55	16.86	-	-
Zero-Shot (MT)	16.66	-	-	16.66	-	-	-	16.66	-	-
ICL Random	32.99	30.68	30.79	33.40	28.02	32.67	33.29	30.64	32.24	31.63
ICL SBERT	33.55	32.84	30.51	37.08	31.85	31.17	29.74	34.62	29.82	33.05
ICL SBERT (MT)	35.80	-	-	37.79	-	-	-	39.19	-	-
X-ICL Random	32.33	28.98	31.12	30.42	33.67	30.39	30.77	30.22	33.50	26.32
X-ICL SBERT										
Source-Only Label	36.99	34.12	34.28	32.93	34.90	32.38	30.57	34.34	32.80	35.49
+ Query Alignment	34.07	34.69	34.29	38.55	31.72	32.85	34.15	31.02	32.94	32.67
Target-Only Label	36.12	-	-	28.67	-	-	32.74	31.57	-	-
Label Alignment	18.41	-	-	17.48	-	-	18.35	20.55	-	-

Table 16: Experiment results for BLOOM-7B1 model on AmericasNLI dataset. "-" denotes the experiment is not conducted due to no machine translation system is available.